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Automated Multi-Hop Dataset Generation for Benchmarking Cited Text Span Extraction in Scientific Question Answering

Bachelor's Thesis

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1. Introduction

Large Language Models (LLMs) have fundamentally transformed how we look up information and obtain answers to questions, becoming an increasingly important tool, not only in everyday life, but also across a range of professional fields. With the use of such systems, the need to verify the factual correctness and easily do further research on the underlying sources arises, especially when answers lack important details or provide a depth of explanation that is misaligned with the user's existing knowledge. This is especially important in high-stakes fields such as medicine, law or scientific research, where seemingly plausible but unbacked or false claims could lead users to draw false conclusions.

One increasingly popular method is the approach of Retrieval Augmented Generation (RAG), replacing zero-shot LLM solutions with a two-step process. Firstly, relevant sources in the form of documents or other multi-media data are retrieved and then fed into the generating model for it to use as ground truth for answer generation and later to enable citation extraction. Retrieving correct and precise text-spans alongside the generated answer is not a trivial task, and with domain-specific requirements widely varying, such systems are part of active research [1] [2]. One system applying such a RAG approach is SQuAI [3], aimed at answering complex open-domain scientific questions on its knowledgebase containing 2.3 million full-text scientific papers. For further performance improvements, it introduces a multi-agent pipeline, enhancing document retrieval capabilities as well as providing document-level backlinks for parts of its generated answer. This system, while providing some backlinking, does still fall short of providing more granular, sentence-level cited text spans, which have proven to be much more useful to users, as they are quicker and easier to verify, enhancing trustworthiness and usability of the system.

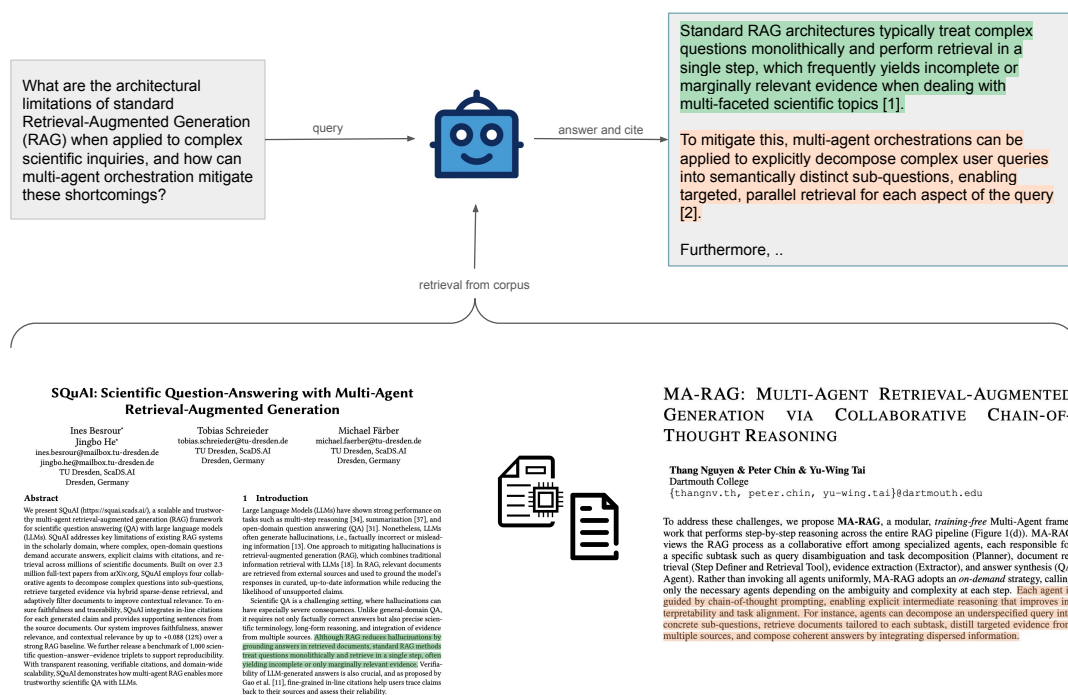


Figure 1.1.: Sentence level cited text span attribution

Therefore, this thesis will focus on benchmarking different cited text span extraction methods to enhance source attribution, leading to the following core contributions:

1. Curate a dataset of high-quality, multi-hop scientific questions building upon a large-scale retrieval corpus that require complex reasoning capabilities in order to be answered and closely resemble real-world scientific inquiry needs.
2. Introduce a hierarchy of different cited text span extraction methods ranging from simple lexical and semantic matching up to complex reasoning capabilities, as well as strategic combinations of such, to allow for tradeoffs in accuracy and efficiency.
3. Apply an automated framework to benchmark the retrieval and answering capabilities of non-parametric, retrieval- and evidence-based text generation systems on the proposed dataset as well as evaluate the extraction methods in terms of efficiency, correctness and quality of source attribution using NLI models and LLM-as-a-judge approaches.

2. Related Work

This chapter provides a comprehensive review of prior work across three key dimensions relevant for the development of a scientific QA system capable of providing cited text spans as evidence for its generated answer. First, it examines existing scientific QA datasets with respect to their suitability for SQuAI. It then compares different established cited text span extraction paradigms and their tradeoffs in terms of precision, efficiency and computational requirements before discussing automatic approaches for evaluating text span attribution quality in long-form, multi-hop settings.

2.1. Datasets

Utilizing SQuAI as a system basis, this work conducts a systematic evaluation of cited text span extraction methods for fine-grained source attribution in scientific QA. Therefore, a highly specialized dataset of open-domain, long form scientific questions is needed. Due to the scarcity and high cost of curating such datasets, only a limited number of suitable options are available. With SQuAI being designed as a system to answer complex scientific inquiries based on the unarXive 2022 Dataset [4], further requirements are introduced, namely:

1. **Domain Alignment:** Questions must fall under the knowledge scope of the underlying knowledgebase, in order for the system to be able to provide cited text spans as proof alongside its answer. In terms of the SQuAI System, this boils down to questions targeting physics, mathematics and computer science that are also part of the papers in the unarXive 2022 Dataset.
2. **High Complexity:** Questions must go beyond simple factoid questions and require complex reasoning capabilities to answer, as questions posed by scientists usually require synthesizing, interpreting and evaluating data, requiring high cognitive abilities to interact with. In order to be able to test if the SQuAI system can be used under such circumstances, the questions in the dataset must also have a similar degree of complexity.
3. **Multi-Hop:** Questions must require the information in multiple documents to be answered, in order to test the ability of the system to synthesize information from multiple sources, as is required in naturally occurring scientific research.
4. **Decontextualization:** Questions must be answerable in an open-domain setting and cannot contain any kind of reference to specific papers and their contents that lose meaning or become ambiguous outside of the context of their paper. As a result, questions must contain enough context such that they are standalone answerable.
5. **Text format:** Questions can only target information in a paper that is present in the form of text, as current versions of the SQuAI system do solely rely on textual information, not taking into account images, graphs or similar.

6. Ground Truth: In order to evaluate the correctness of the model’s answer, a dataset must provide a ground answer alongside each question.

However, comparing some of the most prominent datasets shows that none fits all discussed criteria, finally resulting in the need for a newly curated dataset.

Dataset	Domain Alignment	High Complexity	Multi-Hop	Decontextualization	Text format	Ground Truth	Sufficient Dataset size (>500)
Qasper [5]	partial	shallow	no	no	yes	yes	yes
ALCE [6]	no	shallow	yes	yes	yes	yes	yes
SPIQA [7]	yes	complex	no	partially	no	yes	yes
QASA [8]	yes	complex	no	no	yes	yes	yes
ExpertQA [9]	yes	very complex	yes	yes	yes	yes	no ¹
SciDQA [10]	yes	very complex	no	no	no	yes	yes
PeerQA [11]	yes	very complex	no	no	yes	yes.	yes

Table 2.1.: Comparison of Scientific QA Datasets against SQuAI Requirements

Abbreviations: CS = Computer Science; ML = Machine Learning; NLP = Natural Language Processing.

2.2. Cited Text Span Extraction

Context extraction methods can be most broadly divided into pre- and post-hoc extraction methods, depending on the time the context attribution happens. While pre-hoc systems such as QASPER [5] and QASA [8] extract citable passages before and during runtime of the question-answering system, post-hoc systems search and attach supporting evidence afterwards. Post-hoc systems have been shown to outperform pre-hoc systems, achieving higher citation coverage and answer correctness as well as lowering model hallucination [12]. While a number of systems rely on zero-shot LLM solutions for context extraction [13] [14] [15], work on the DECOMPTUNE [16] system has shown that decomposing facts and decontextualizing answer sentences before attributing source text spans with LLM prompting can yield significant improvements, especially when applied to answers for long-form, multi-hop questions. In order to eliminate the probability of hallucinated model output, models can be finetuned [17] or the answer space restricted to the start and end token of the cited text span, forcing strict citation [**citationContextExtractionUsingATransformerBasedModel**].

So-called white-box methods rely on access to the model’s internal weights, rendering them incompatible with proprietary LLMs. The LAQuer [18] system, as well as earlier token-embedding approaches it is built upon [19], analyzes internal hidden states to extract the most likely text span that a user-highlighted part of the LLM’s answer is based upon. To pinpoint a certain text span, the system calculates the maximum cosine similarity between individual tokens in the source text and the averaged hidden representation of the highlighted span, for which attribution is sought. The system then extracts the precise quote from the surrounding window of text. However, this method has been shown by the authors to perform significantly worse than LLM Prompting, with drastically lower accuracy and missing suitable citations, especially in larger search spaces, e.g.

¹question-answer pairs are broadly distributed across diverse fields of study, resulting in insufficient topical overlap with unarXive 2022

longer text. Similarly, AttnAttribute [20] identifies "Attention Heads", tracking the model's internal attention values during generation time. Just like LAQuer, it then extracts text spans from the passage surrounding the anchor with the highest model attention. Benchmarks made as part of the development of DECOMPTUNE [16] concluded that both of those methods performed significantly worse than LLM Prompting.

Recent advancements have explored generative approaches. Rather than retrieving and reranking text spans, such systems generate them. In order to prevent hallucinations, the output vocabulary of the systems is typically restricted to the underlying retrieval corpus by applying constrained decoding mechanisms, such as prefix trees or FM-indices [21]. GERE [22], as one such system using a prefix tree, applies this concept of restricted output in the document retrieval step, generating document titles for selecting appropriate source texts, as well as sentence identifiers mapped to individual sentences. This system, while removing the need for a massive document index as well as showing significant improvements in precise and correct text span extraction compared to baselines such as Dense Passage Retrieval (DPR) and BM25, requires the prefix tree to be precomputed and stored in memory, making it highly inflexible for changes in the underlying corpus. Similarly, MRR-FV [23] further improves this approach with multi-hop reasoning capabilities. Instead of generating sentence identifiers, it generates sentences directly, leveraging the natural language understanding capabilities of LLMs, leading to improvements in precision and quality of attributed spans, but also further bloating the prefix tree, making the system even harder to scale.

Despite the diversity of proposed paradigms, current work mostly evaluates them in isolation, restricting comparison to baselines or direct predecessors on heterogeneous tasks, making broad comparisons for practical applicability unfeasible. Consequently, this work introduces a hierarchy of extraction methods, increasing in both reasoning capabilities and computational requirements, benchmarking both accuracy and efficiency on a unified framework, allowing for systematic trade-off analysis between paradigms in the context of open-domain multi-hop scientific QA.

2.3. System Evaluation

While human annotation has always been the gold standard for assessing natural language generation (NLG) tasks, relying solely on it is a bottleneck for developing models of increasing size and complexity. While providing feedback with extensive reasoning and natural language understanding, human evaluation suffers from being costly, time-intensive and, depending on the specific task, highly subjective. Thus, lowering or even replacing the need for human evaluation completely is an important problem.

Early metrics heavily rely on measuring the lexical overlap between text spans, for instance of a human-written ground truth and an LLM-generated answer for the same questions. The most prominent examples include BLEU [24] and ROUGE [25], both being n-gram based metrics. While the former is precision-oriented, measuring how many of its n-grams also appear in the human reference span, the latter applies a recall-oriented approach, measuring how many n-grams of the human reference are also present in the LLM-generated text span. Being computationally inexpensive and language independent, these metrics have established themselves as rigid industry standards and

baselines for other methods. Their alignment with human judgement, however, has been shown to be low, suffering from the lack of semantic understanding, leading to penalizing valid paraphrasing as well as missing contradictory or hallucinated content.

Partly overcoming those limitations, embedding-based metrics like BERTScore [26] and MoverScore [27] emerged, measuring semantic instead of lexical similarity. BERTScore does one-to-one greedy matching on vector embeddings, providing precision, recall and F1 score metrics, just like BLEU and ROUGE, but on an embedding level. MoverScore improves upon this idea, utilizing Earth Mover’s Distance (EMD) to allow for introducing one-to-many relationships, allowing for matching of single words with phrases, so-called soft matching, as well as introducing Inverse Document Frequency (IDF) for enhanced lexical matching, to better reflect the importance of specific keywords to be included. Though significantly more expensive to compute, those metrics show significant improvements in semantic robustness as well as intrinsic context awareness, due to their vector embeddings. However, both of those metrics still require a reference answer to compare to. BartScore [28] eliminates this requirement, measuring the conditional probability that one text span is generated from another using an encoder-decoder model, arguing that the probability is higher if the claims in the answer are also deeply rooted in the source text.

Besides similarity-based metrics, other research suggests approaching the task from a logical entailment perspective. The AIS [29] framework proposed a unified conceptual definition for such, arguing that a statement is considered Attributable to Identified Sources (AIS) iff completely corroborated by a source. In order to verify this, the authors imposed additional constraints being 1) that the claim made in an answer sentence must be standalone interpretable, thus decontextualized and 2) the information in the text span must support all the information present in the answer span it is attributed to. Previously discussed similarity-based metrics like BLEU, ROUGE or token-level F1, as well as solely relying on an out-of-the-box Natural Language Inference (NLI) model, have been shown to have low correlation with human judgement for both factual consistency and logical entailment assessment [30] [31]. To address these limitations, systems relying on NLI models like FactCC [31] argued that evaluating logical entailment on sentence-to-sentence level is insufficient, since information in sources might be spread across multiple passages. As a result, these systems increased their evaluation scope to the whole source text, leading to better judgement capabilities, but also increase in model input. The authors of FactCC observed that a majority of misclassifications were commonsense mistakes; the TRUE benchmark [30] observed improvements with increasing model size but also degradation with longer input, which is further supported by research on the SummaC [32] framework. To shrink input size, SummaC increased evaluation granularity by dividing both the source text and the model answer into smaller units, checking each combination of them for entailment and combining the obtained scores.

The Q² system [33] proposed using a Question-Generation - Question-Answering (QG-QA) method, generating questions to which the LLM-generated answer is the solution. Those questions are then answered with the original source document and lastly the so obtained answers are then compared with the original LLM answer using an NLI model. The authors presented significant improvements compared to baseline methods, further enhanced by the TRUE benchmark by applying an ensemble approach utilizing ANLI (standard NLI), SummaC (increased granularity NLI) as well as the Q² (QG-QA) method. The ensemble was shown to beat every standalone evaluation metric previously

discussed on all datasets benched in TRUE, emphasizing the need for combined, complementary evaluation metrics.

Recently, the evaluation paradigm has shifted to LLM-as-a-judge approaches. While early versions like GPTScore [34] tried evaluating based on model internals, using the conditional token probability, thus the confidence of a model in a given answer, they have quickly been shown to be much less effective than prompting the LLM directly to grade [35]. The performance of LLMs in evaluation tasks can additionally be enhanced by applying a Chain-of-thought approach, prompting the model to break down complex tasks into simpler ones and adding intermediate reasoning steps. Closely resembling human cognitive processes, this method yields significant improvements, especially with larger models and problems of increasing complexity [36], reaching agreement rates comparable to human judges for open-ended multi-turn dialogues [37], thus eliminating human annotation constraints.

When applied to long-form text containing multiple pieces of information, binary judgements become insufficient as they cannot reflect partial attribution. FactScore [38] addresses this by breaking down answers into atomic facts using an LLM, obtaining a percentage measure of supportedness. AttrScore [39] introduces additional evaluation criteria, such as measures for attribution, extrapolation and contradiction, in order to further characterize the type of attribution error. Similarly, CiteEval [40] introduces penalties for attributions containing unnecessary noise and redundant citations while advocating for evaluation against full source documents, in order to not lose information in the surrounding context. The RAGAS [41] framework operationalized these findings, providing a suite of automated metrics like faithfulness, answer relevance and context relevance, without requiring human annotation. However, the importance of penalizing undesired behaviors is further emphasized by the MT-Benchmark [37], showing that LLMs have a multitude of biases, such as favoring longer and LLM-generated text spans or switching preferences between answers when presented in a different order. Furthermore, LLMs perform remarkably poorly with fine-grained information, with a majority of mistakes being caused by overlooking numbers and entities or hallucinating logical connections [42]. Using increasingly large models additionally introduces high computational cost requirements. To address this, systems like ARES [43] rely on LLM-generated, domain-specific, synthetic training data to finetune smaller models. Using Prediction-Powered-Inference (PPI), ARES leverages a small human-annotated dataset to calculate the LLM’s annotation error on the synthetic data, the "rectifier", to provide statistical confidence bounds and enhance its evaluation, beating existing zero-shot solutions like RAGAS.

Automatic evaluation of cited text span attribution remains a challenging task. Despite the wide range of approaches, no single metric being capable of reliably and comprehensively assessing attribution performance as a whole has emerged. Increasingly complex tasks, such as multi-hop questions and challenging application domains like scientific QA, introduce further evaluation dimensions. Consequently, combining multiple evaluation metrics with complementary strengths becomes necessary to measure task-specific aspects within a unified framework.

3. Methodology and Approach

3.1. Dataset

To curate a dataset of high-quality, expert-level scientific question-answer pairs, a fully automated pipeline is applied. It consists of three consecutive steps: document selection, question-answer generation and quality assurance in order to ensure that generated questions and answers are scientifically rigorous and strictly dependent on multi-document information synthesis in order to be answered, while also reliably providing associated ground-truth documents alongside them.

3.1.1. Document selection

To match the underlying document corpus, selectable papers are restricted to those of the unarXive 2022 dataset, yet for other applications, this could be expanded at will. A randomly selected anchor paper as well as its cited references serve as the foundation for generating a single question-answer pair. For each paper, a document-level SPECTER [44] vector embedding is retrieved via the Semantic Scholar academic graph API [45]. Leveraging citation graphs, these vectors provide dense representations of a paper's title and abstract, which are subsequently used to calculate the cosine similarity and thus the semantic similarity of each paper with the anchor document. To ensure topical alignment while excluding highly redundant work, a subset of candidate papers with a cosine similarity between 0.7 and 0.9 is selected. Four of these papers are then chosen at random to be used alongside the anchor paper for generation. Restricting the model input to five papers in total provides a multitude of topical overlaps for the model to choose from without resulting in exceedingly long prompts, mitigating the risk of "Lost in the Middle" effects. If a chosen anchor paper fails to provide at least four appropriate references, it is considered insufficient to be used in the dataset.

3.1.2. Question-Answer generation

While modern LLMs offer increasingly large context windows, usually even above 100,000 tokens, empirical studies have shown severe performance degradation in tasks like complex reasoning, data retrieval and logic problems, accompanied by drastically increased hallucination [46]. In the generation step, the pipeline applies a moonshotai/Kimi-K2.6 [47] model, supporting a context window of approximately 256,000 tokens. In practice, however, the "effective context window", in which models only suffer from minimal performance degradations with increasingly long inputs, has been shown by empirical studies [48] and practitioner reports to be drastically lower, with some studies estimating it as low as 40% of the maximum supported context window [49]. To ensure high reasoning capabilities, the pipeline applies an adaptive character limit for each paper text, enforcing a total upper bound of 200,000 characters for all papers combined, which conservatively translates to roughly 66,000 tokens for scientific paper texts [50], or about 26% of Kimi-K2-6's supported

context size. Papers are assigned input space according to the following formula, allowing for full and efficient utilization of the context window, even if papers are highly variable in length:

$$C_{adaptive} = \left\lfloor C_{base} + \left(\frac{l_{paper}}{\sum_{p \in P_{above}} len(p)} \right) \times \sum_{s \in P_{below}} (C_{base} - len(s)) \right\rfloor \quad (3.1)$$

Paper texts longer than their adaptive character limit are then head- and tail-truncated, retaining the first and last $\lfloor \frac{C_{adaptive}}{2} \rfloor$ characters, which are the most information-dense parts of scientific papers.

Subsequently, the LLM is prompted using a two-step strategy. Firstly, the model is tasked with selecting the best pair of papers (A, B), where paper A provides an intermediate entity, method, dataset or a similar element, and paper B evaluates, extends, contradicts or relies on A’s intermediate element. Following that, the LLM is tasked with generating a single question-answer pair based on its selected pair, whose answer can only be derived by following a sequential reasoning process across both papers, applying paper A’s intermediate element to paper B. Additionally, the model is instructed to answer the question as precisely as possible, providing a precision-biased subset of the underlying papers used during generation alongside.

3.1.3. Quality assurance

In order to ensure the dataset’s integrity, two separate evaluation steps are applied, both using an openai/gpt-oss-120b [51] model. The first judge evaluates each Question-Answer pair along six distinct dimensions, providing a judgement on how well each criterion is satisfied on a three-point ordinal scale. The complete prompt instructions for each, are provided in the appendix.

1. **Multi-Hop Validity** assesses whether answering the question requires integrating information from both papers. Thus, this criterion is only satisfied if neither paper provides sufficient information for deriving the complete answer. A question only satisfies this criterion if neither paper alone provides sufficient evidence to derive the complete answer (A.2).
2. **Dependency Strength** evaluates whether the steps in the reasoning chain are sequentially dependent in order for the conclusion to be drawn. Thus, the second reasoning step cannot be solved without first applying or interpreting the first one (A.3).
3. **Non-Decomposability** verifies that questions cannot be broken down into independent single-hop sub-questions, allowing for reasoning shortcuts or simply combining sub-question answers. Instead, reasoning steps must be connected in such a way that separately solving them would not be sufficient for deriving an answer (A.4).
4. **Evidence Distribution** examines whether both papers contribute distinct and necessary evidence, ensuring that required evidence is spread across both papers and that questions cannot be answered by using a single dominant source with redundant information (A.5).
5. **Answerability** assesses whether the answer is fully supported and obtainable by only relying on information present in the papers, without requiring external knowledge or speculative

inference not explicitly supported by either paper. Is the answer fully supported by the provided evidence (A.6)?

6. **Decontextualization** validates that the question is self-contained and unambiguous in an open-domain setting. Thus, questions cannot contain context-specific references and must instead introduce relevant concepts, methods or entities by themselves in order to satisfy this criterion (A.7).

An analysis of disconnected reasoning by Trivedi et al. [52] demonstrated that many alleged multi-hop question-answering datasets do in fact not require true multi-hop reasoning, but can instead be solved through reasoning shortcuts, partial evidence or decomposition into independent sub-questions. Using a methodology inspired by the DIRE [52] framework, the second judge verifies that questions cannot be answered under incomplete evidence, namely a subset $P' \subset P$ of ground truth papers P , but instead require all ground truth documents for a complete reasoning chain and answer. Thus, the test outcome satisfies:

$$(\forall P' \subset P : \neg \text{Pass}(P')) \wedge \text{Pass}(P)$$

The final dataset is then obtained by filtering question-answer pairs to only those successfully passing both judges. In order to verify the pipeline and the judges, human verification will be applied on a subset.

3.2. Cited text span extraction

Constrained by the SQuAI system’s attribution granularity, which natively provides exactly one source document per answer sentence, the source span extraction task in this work is formulated as a post-hoc attribution step for providing one source span per answer sentence. The retrieval methods themselves, however, are applicable to other systems with alternative attribution granularities. This constitutes a limitation for this specific system, especially in cases where one attribution scope (e.g., a sentence) contains multiple claims requiring evidence, but also avoids additional ambiguity introduced by finer attribution granularity. For the purpose of comparing different extraction methods under identical conditions and constraints, however, this granularity is sufficient in order to measure whether extraction variants are capable of identifying relevant evidence, with results being directly comparable.

As previously discussed, cited text span attribution can be approached through a number of different paradigms, with methods being introduced and applied under vastly different circumstances. In the following, a hierarchy of increasingly complex post-hoc extraction methods that will be applied on a unified dataset is introduced. All methods except the native SQuAI span extraction and the generative LLM method receive tuples composed of an answer sentence and a set of eligible source spans $\mathcal{W}(P)$ from the paper P , annotated as the factual basis for that sentence during question answering. The set of source spans is composed of floating windows of contiguous sentences in the source document, up to a length of five, and can be denoted as:

$$\mathcal{W}(P) = \bigcup_{k=1}^5 \{s_i \oplus s_{i+1} \oplus \dots \oplus s_{i+k-1} \mid 1 \leq i \leq n - k + 1\},$$

where k specifies the window size and $\oplus$ denotes string concatenation with whitespace.

Extraction Method	Floating windows	Retrieval Category	Candidate Selection (Top 10)	Final Selection (Top 1)
1	×	Native	-	SQuAI
2	✓	Dense	-	Bi-encoder
3	✓	Sparse	-	BM25
4	✓	Hybrid	Bi-encoder	BM25
5	✓	Hybrid	BM25	Bi-encoder
6	✓	Hybrid	-	BM25 & Bi-encoder ²
7	✓	Cross-encoder	Bi-encoder	Cross-encoder
8	✓	Cross-encoder	BM25	Cross-encoder
9	✓	Cross-encoder	BM25 & Bi-encoder ²	Cross-encoder
10	×	Generative	-	LLM

Table 3.1.: Cited text span extraction methods overview

3.2.1. Native SQuAI extraction baseline

The first method relies on SQuAI’s native text span extraction mechanism, utilizing simple lexical matching. Since it is only capable of providing one span per paper, even if it is cited repeatedly in different contexts, it is limited by design. Nevertheless, representing the attribution behaviour of the SQuAI system, it serves as a native baseline for further extraction pipelines to compare against.

3.2.2. Dense retrieval

Applying a bi-encoder, this method selects the best-fitting source span out of all floating windows based on the similarity of the span’s dense vector representation with that of the answer sentence it is attributed to. It measures the suitability and sufficiency of relying exclusively on semantic similarity for source span attribution. As a dense retrieval baseline, it is capable of identifying appropriate source spans with different wording, but is prone to overlooking precise terminology.

3.2.3. Sparse retrieval

The third method utilizes BM25, reranking source spans based on their lexical similarity with the corresponding answer sentence. The highest-scoring span is then selected as the final attributed

² using Reciprocal Rank Fusion

span. Representing the sparse retrieval baseline, BM25 is especially effective with precise terminology, but less robust to paraphrasing. Its purpose is to test whether focusing on specific terms, entities or important keywords alone proves effective for cited text span extraction.

3.2.4. Hybrid retrieval

Hybrid methods aim to combine sparse and dense retrieval, leveraging their distinct characteristics. Thus, these variants evaluate whether the retrieval performance can be enhanced by combining both approaches into a single method. The fourth method applies a two-stage retrieval method, with the bi-encoder selecting the top ten candidate windows and BM25 selecting the final span, while method five reverses their order. While the former prioritizes semantic recall during candidate selection and lexical precision during final span selection, the latter reverses this order. Comparing both allows for analysis of whether either of those paradigms is better suited for candidate or final span selection. The sixth method combines both lexical and semantic matching into a single score using Reciprocal Rank Fusion (RRF). Instead of sequential application, this variant evaluates hybrid retrieval performance without an imposed order between the components themselves. Thus, this combination allows for assessment of whether the paradigms capture complementary aspects of the attribution task, resulting in performance enhancements when applied at the same time.

3.2.5. Cross-Encoder reranking

Overcoming the limitation of BM25 and bi-encoder retrieval processing the answer sentence and source span in isolation, the following methods introduce cross-encoders, allowing for joint processing of both. This promises improvements, especially for context windows where fine-grained semantic differences such as negations or conditions as well as important entities or numerical values determine whether the answer sentence is merely related, contradicted or truly supported. These variants therefore evaluate whether joint answer-span comparison translates into better evidence selection in the form of cited text spans. However, as cross-encoders are far more computationally expensive than previous methods and thus not feasible to run on the whole set of floating window candidate spans, these methods instead use different candidate selectors, with the cross-encoder selecting the best-fitting final span for the answer sentence. Methods seven and eight apply semantic and lexical matching, respectively, while method nine utilizes both together as a candidate selector using RRF. Comparing these variations allows assessing which candidate selection strategy provides the best basis for a cross-encoder and therefore improves overall attribution quality.

3.2.6. Generative LLM

The last method uses a generative LLM to extract the best supporting span from the source document. Unlike previous methods, it does not rerank candidate spans, but instead prompts the model to identify the best-suited text span directly. Representing a generative approach, it is used to test whether an LLM’s reasoning capabilities can improve evidence selection beyond retrieval and reranking, serving as a high-cost reference method.

3.3. System evaluation

The performance of the text span extraction methods can only be evaluated conditionally on the outputs of preceding pipeline steps, since post-hoc attribution is heavily dependent on their results. The evaluation is therefore split into assessing the suitability of the upstream inputs for text span extraction and evaluating the quality of the extracted spans themselves. This separation prevents text-span evaluation results from being confounded by errors introduced before span extraction.

3.3.1. Upstream system evaluation

Upstream inputs leading to bad span attributions can be flawed for a number of reasons, including:

1. Retrieval failure: The system retrieves or filters the wrong or insufficient subset of papers.
2. Generation failure: The system generates a factually wrong, overgeneralized, hallucinated or insufficiently supported answer.
3. Document attribution failure: The system fails to attribute or incorrectly attributes source documents to answer sentences.

These must be addressed separately when evaluating the attributed spans.

Retrieval failure can be analyzed by comparing the retrieved documents with those used during question generation, making it possible not only to evaluate the retrieval capabilities of the SQuAI system itself using precision, recall and F1-scores, but also the conditional performance of attribution methods given only unrelated, partial or complete evidence sources.

Generation failure is examined by comparing the system-generated answer with the ground-truth answer provided in the dataset. This comparison evaluates whether the system’s answer covers at least the factual content of the reference, including all relevant facts and without introducing contradictory information. Exact lexical overlap is not required, as the comparison is performed by an LLM-as-a-judge, whose reliability is validated through human annotation on a small subset.

Document attribution failure can be evaluated by running extraction methods on all retrieved source documents, regardless of the sentence-specific document annotation by the answering LLM. This allows for comparing the best spans from both the annotated document and the whole set of retrieved papers. If no sufficient span can be extracted from the former, while the latter provides one, the failure indicates incorrect document attribution rather than a failure of the span extraction method itself.

3.3.2. Span evaluation

The attributed spans are evaluated across multiple distinct dimensions in span-sentence pairs, with the goal of assessing whether they provide sufficient evidence for claims made in their respective answer sentence. This evaluation applies both an LLM-as-a-judge approach with RAGAS-inspired metrics, such as faithfulness, answer relevance and context relevance [41], as well as an NLI model to check for logical entailment between the span and the answer sentence. Additionally, the runtime required to extract spans will be compared across methods, allowing the evaluation to capture the trade-off between attribution quality and computational cost.

4. Evaluation

TODO write intro for evaluation

4.1. Dataset

4.1.1. Dataset filtering

The dataset generation process attempted to construct question-answers pairs from anchor papers, out of which 11203 provided enough cited papers with similar or related content, measured by the SPECTER vector embeddings, resulting in successful generation. Out of those candidates, 470 ($\sim 4.2\%$) were retained for the final dataset, scoring a "GOOD" judgement for each quality assurance category such as Multi-Hop Validity or Non-Decomposability, as well as passing the DIRE filter; both methodologies are described in section 3.1.3. While this highly restrictive approach may also discard potentially good questions, this dataset focuses on precision rather than recall in order to ensure that subsequent evaluations are undistorted, thus aiming to minimize the number of unsuitable QA-pairs in the filtered dataset. The resulting judgements of the quality assurance categories and DIRE filter are distributed as follows:

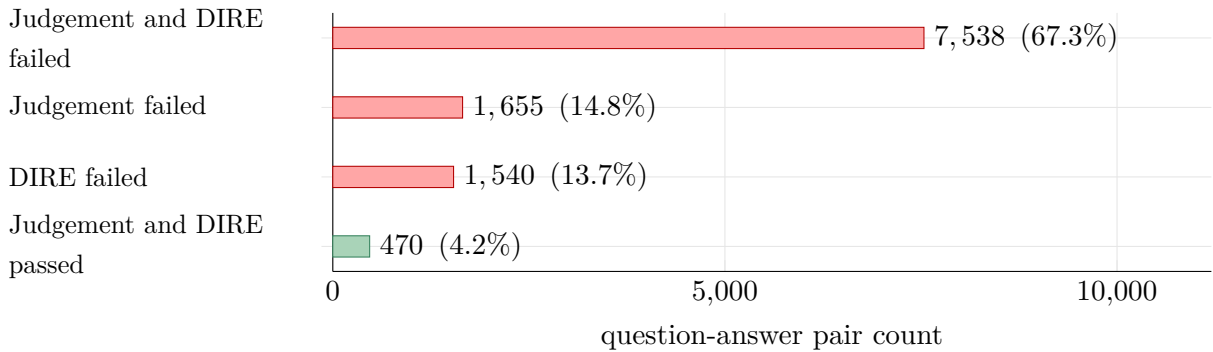


Figure 4.1.: Quality-assurance and DIRE filter outcomes

While the majority of questions satisfy neither all quality-assurance criteria nor the DIRE filter, a substantial share of 28.5% passes exactly one of the two filtering steps. While both the Multi-hop criterion and the DIRE filter are intended to identify insufficient multi-hop structure in the questions, the former is obtained by directly prompting an LLM and showed to be much less effective than the DIRE filter. A closer comparison reveals, that the rejected subsets of questions are overlapping but not identical, with DIRE rejecting almost twice as many as the Multi-hop criterion in absolute numbers. However DIRE rejects almost all candidates that failed the Multi-hop criterion but not vice versa, supporting the complementary usage of both filtering steps.

$$P(MH_{\text{fail}} \mid DIRE_{\text{fail}}) = (n_{\text{each}} \cdot p_{\text{each}} + n_{\text{exactlyOne}} \cdot p_{\text{exactlyOne}} + n_{\text{unanswerable}} \cdot p_{\text{unanswerable}}) / n_{DIRE_{\text{fail}}} \approx 46.5\%.$$

$$P(DIRE_{\text{fail}} | MH_{\text{fail}}) = (n_{\text{each}} \cdot p_{\text{each}} + n_{\text{exactlyOne}} \cdot p_{\text{exactlyOne}} + n_{\text{unanswerable}} \cdot p_{\text{unanswerable}}) / n_{MH_{\text{fail}}} \approx 90.9\%.$$

Figure 4.2 shows the distribution of quality-assurance criteria failures. Since they are not mutually exclusive, a candidate QA-pair might fail several at once. Non-Decomposability accounts for the largest number of failures, closely followed by Multi-Hop validity failures, while Answerability represents the third largest failure category. Together, this suggests that the biggest problem in the generation process remains the construction of a genuinely connected reasoning chain across both papers, including distinct information from both papers, while also not overgeneralizing or -interpreting said information, so that the question itself is still grounded in the source documents. This argument is further supported by the low conditional co-failure of the Evidence Distribution criterion as shown in figure 4.1, suggesting that while both papers do contribute distinct evidence, the LLM is unable to utilize it for the creation of a grounded, Non-Decomposable multi-hop question. Evidence Distribution failures themselves can be caused in two ways, by either having no meaningful overlap in the five papers presented to the LLM or it being unable to extract them. While not as low as decontextualization failures, this also remains a rare problem, suggesting that the LLM is quite capable of extracting them. In contrast, the LLM shows strong capability of decontextualizing the information, with errors caused by non standalone answerable questions arising so scarcely, that they are practically irrelevant.

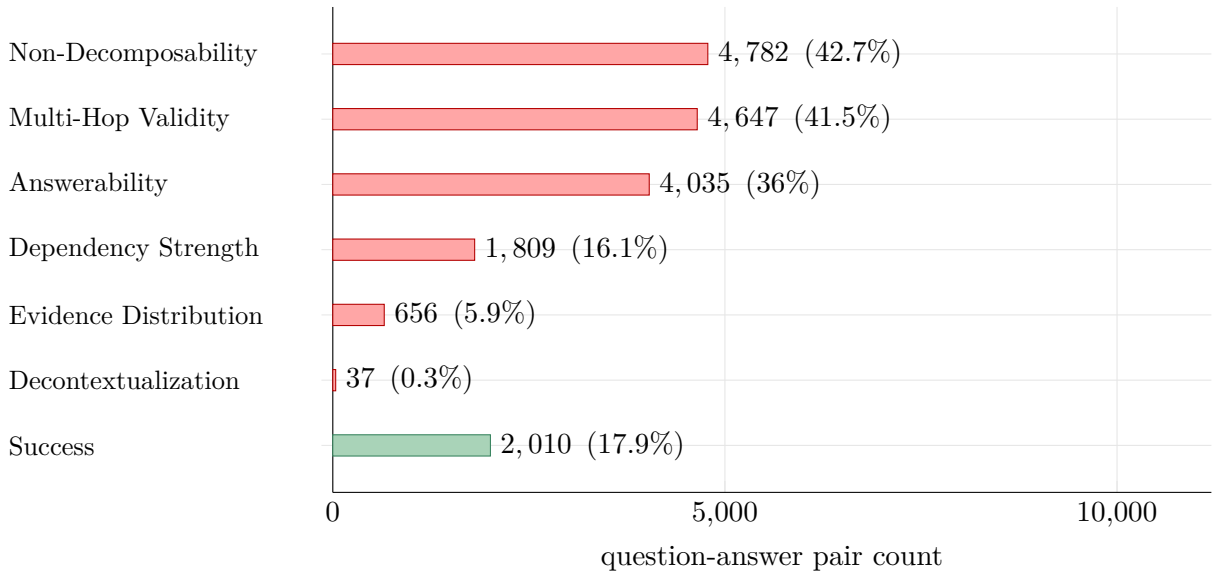


Figure 4.2.: Quality-assurance outcomes

Figure 4.3 shows the mutually exclusive outcomes of the DIRE filter. By far the most prominent failure is the answerability of a question using exactly one of the two provided papers, indicating that a majority of generated questions rely on a single dominant source, marking a failure of generating a reasoning chain with genuine cross-paper connection. About 15.4% of questions are answerable using either of the provided papers, being an even worse failure of such construction, targeting information that is redundant in both papers. In about 8.5% of cases, questions are unanswerable even when using all provided papers as evidence, hinting at overgeneralization or hallucination by

the LLM during the generation process. The remaining minority of 19% of questions pass the DIRE filter successfully, again highlighting the precision-oriented character of the generation pipeline.

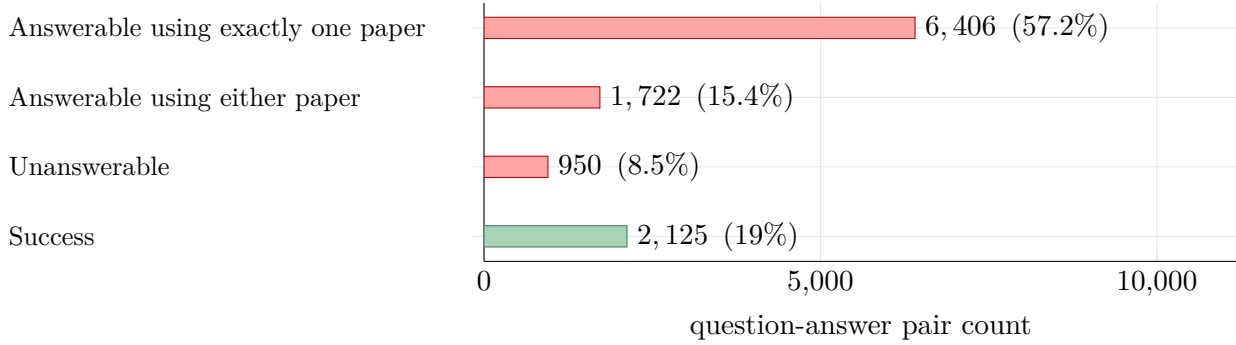


Figure 4.3.: DIRE filter outcomes

given failure	answerability	decontextualization	dependencyStrength	evidenceDistribution	multiHopValidity	nonDecomposability	answerableByEach	answerableByExactlyOne	unanswerable
answerability	-	0.52%	17.60%	3.47%	32.71%	39.43%	9.14%	50.51%	17.32%
decontextualization	56.76%	-	13.51%	5.41%	40.54%	40.54%	5.41%	59.46%	16.22%
dependencyStrength	39.25%	0.28%	-	5.20%	52.24%	48.26%	8.46%	55.89%	11.11%
evidenceDistribution	21.34%	0.30%	14.33%	-	90.09%	54.42%	26.68%	65.85%	3.20%
multiHopValidity	28.41%	0.32%	20.34%	12.72%	-	47.13%	20.42%	66.19%	4.24%
nonDecomposability	33.27%	0.31%	18.26%	7.47%	45.80%	-	14.97%	59.20%	6.98%
answerableByEach	21.43%	0.12%	8.89%	10.16%	55.11%	41.58%	-	-	-
answerableByExactlyOne	31.81%	0.34%	15.78%	6.74%	48.02%	44.19%	-	-	-
unanswerable	73.58%	0.63%	21.16%	2.21%	20.74%	35.16%	-	-	-

Table 4.1.: conditional propability of co-failure. Rows denote the given failure, columns denote the conditional propability under that

4.1.2. Dataset characteristics

Figure 4.4 shows the category distribution as well as its change during different stages of the dataset construction, including the initial random paper selection, approximating the distribution in the underlying corpus, gathering and similarity filtering of cited references as well as the filtering imposed by both the Quality-assurance metrics and the DIRE filter. While the underlying corpus has categories assigned to individual papers, the following distributions must be interpreted on paper pairs instead, due to the intended multihop character of the questions. Therefore a "Interdisciplinary" category is introduced, if both papers don't share the same category.

The observed shifts in percentage share indicate that the pipeline introduces a category-dependent bias, with differing effects across both categories and stages. While Physics is favored by the generation pipeline, increasing its share in both stages, Mathematics and Other get drastically reduced, as

compared to the corpus distribution. Both Computer Science as well as Interdisciplinary questions are not as affected, both lose a bit of share due to the Quality-assurance and DIRE filter.

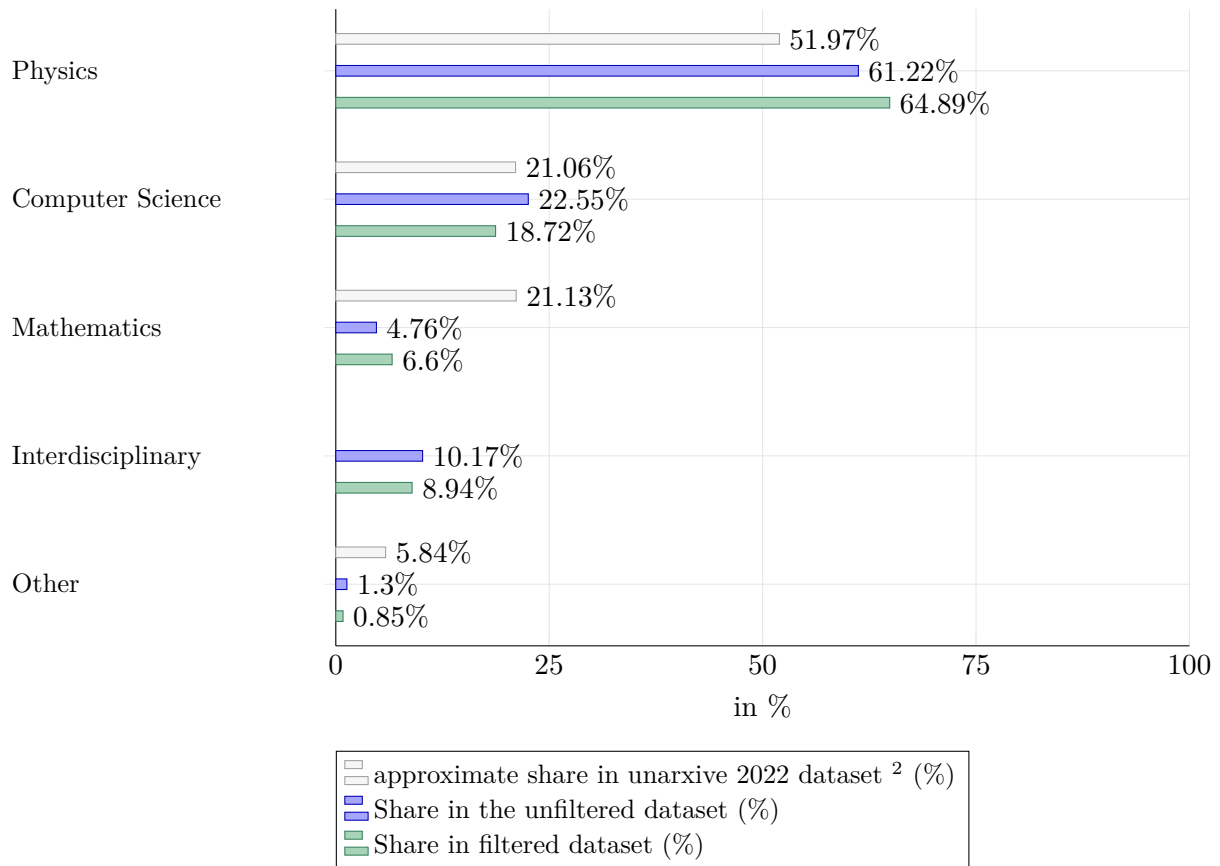


Figure 4.4.: Dataset category distribution

A paper is considered eligible for question generation if it has 1) at least four cited references that are also part of the underlying corpus and 2) at least four of those references are inside the cosine similarity threshold of being greater than 0.7 to ensure overlap but less than 0.9 as to exclude duplicates. Figure 4.5 groups the failure cases by category. With respect to the first criterion, Physics gets least affected, while all other categories fail to provide at least four cited references substantially more often, with the strongest failure rates emerging for Computer Science. A possible explanation for this observation might be, that Physics has by far the largest share in the underlying corpus, therefore referenced papers are also more likely to be included, while scarcely represented categories, especially the consolidated "Other" with the second highest failure rate might be more prone to failure. Although both Computer Science and Mathematics have a substantial share in the corpus, these papers might not have their references included, leaving gaps in the citation graph, resulting in higher failures for those categories as well. This however, remains entirely hypothetical, since the analysis does not include in-depth analysis of category corpus share and corpus-contained reference availability. The second criterion, filtering by cosine-similarity, shows a more balanced distribution. Physics and Mathematics have slightly increased error rates, compared to the average, while both Computer Science and papers in the Other category seem to be less affected, providing similar papers about 7% more often than the average.

²category distribution is approximated by sampling the categories of all randomly selected papers tried for generation

Overall, Physics has the highest eligibility rate for further generation,

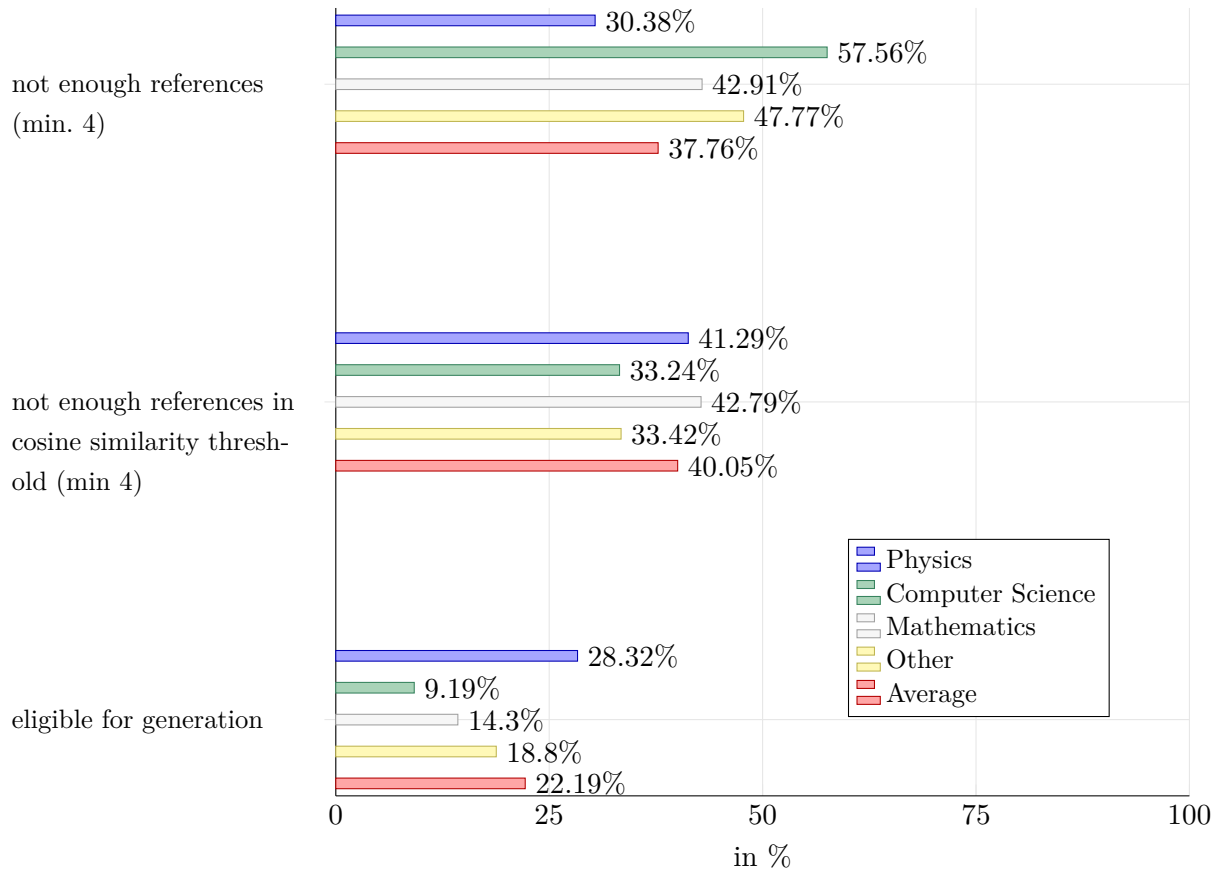


Figure 4.5.: cause of error during generation by paper category

Table 4.2 shows the deviation of category specific judgements from both the quality assurance metrics as well as the DIRE filter. Although the total rejection rate varies only slightly, categories are affected differently by certain parts of the filter.

overall		Physics	Computer Science	Mathematics	Interdisciplinary	Other
82,06%	bad Judgement	-0.12%	1.47%	-4.20%	0.12%	-5.35%
41,48%	multiHopValidity	3.22%	-5.85%	-3.21%	-4.25%	-5.18 %
16,15%	dependencyStrength	-0.14%	1.71 %	-1.51%	-2.28%	0.29 %
42,68%	nonDecomposability	-0.43%	2.17 %	0.09 %	-1.16 %	-8.44 %
5,86%	evidenceDistribution	0.37%	-1.07%	0.90%	-0.32 %	0.31 %
36,02%	answerability	-2.98%	6.54 %	-4.31%	5.25 %	1.65 %
0,33%	decontextualization	-0.01%	-0.01%	0.23%	0.02 %	-0.33 %
81,03%	bad DIRE	0.25%	-0.27%	-4.48%	0.62%	4.58%
8,48%	unanswerableByPapers	-0.53%	0.86 %	-0.22%	0.74 %	5.22 %
15,37%	answerableByEachPaperIndividually	1.16%	-2.62%	0.76%	-1.15 %	-3.04 %
57,18%	answerableEitherAorB	-0.38%	1.49 %	-5.02%	1.03 %	2.41 %
95,80%	bad total	-0.25%	0.71%	-1.62%	0.51%	1.46%

Table 4.2.: deviation of the failure categories relative to the average. Negative values denote less failures, positive values indicate increased failure rates

4.1.3. Manual validation

In order to validate the robustness of the automatic filtering process, a randomly selected sample of 50 computer science questions out of the 470 retained question-answer pairs as well as their filtering results were manually inspected. Caused by the unavailability of domain expertise, the restriction to computer science poses a limitation for generalizing the evaluation results across different fields of study, however, the high manual agreement within this subset suggests, that the filtering pipeline could show similar performance elsewhere. Since this analysis is limited to positive QA-pairs, its effectively measuring filtering pipeline precision rather than recall by conducting manual evaluation on both the quality-assurance criteria, as well as the DIRE filter. The results show high agreement rates from 90% to 96% on individual quality-assurance criteria, while simultaneous agreement on all is measured at 82%. Analysis of the DIRE filter produces even higher agreement rates, with all three instances of misaligned judgements being, that the question is answerable by using one paper only. Overall the manual validation shows 82% agreement with the automatic filtering pipeline, showcasing, that even with its restrictive character a smaller amount of noise remains present in the dataset.

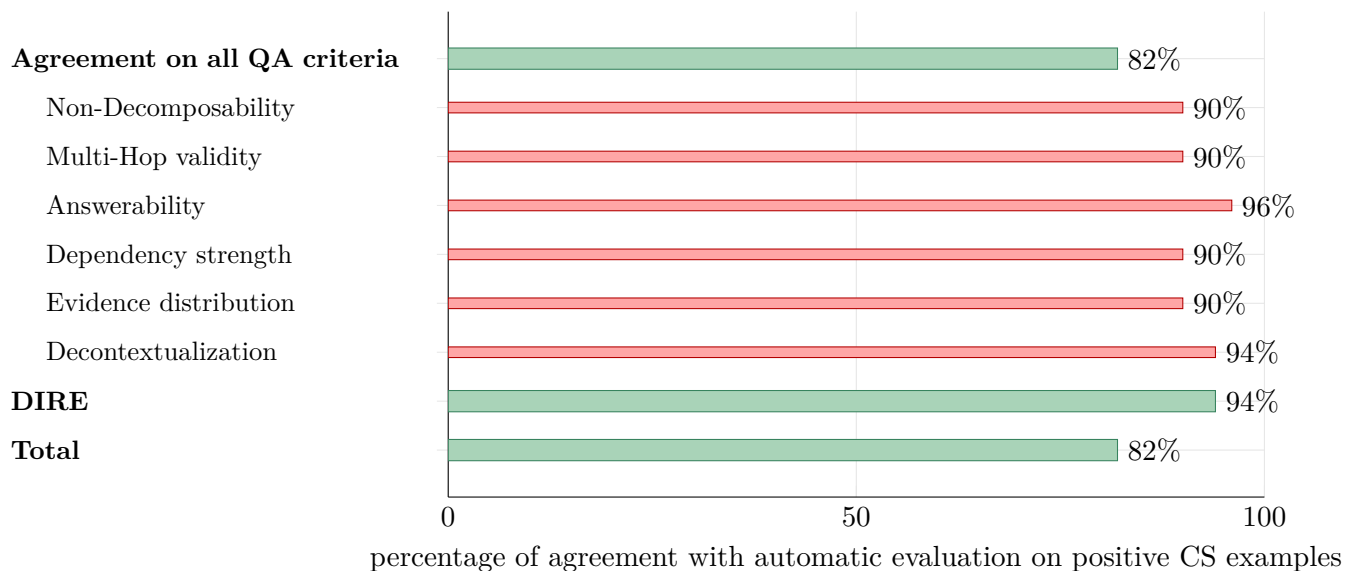


Figure 4.6.: Manual validation

4.2. Retrieval Performance

With the RAG approach, both question answering and cited text span extraction are inherently constrained by the performance of the underlying retrieval component. Figure 4.7 shows the retrieval recall of the SQuAI system on the proposed dataset by category. Across all those, on average, the system retrieves 0.63 (31.7%) of the two ground-truth papers used for question generation. Computer Science once again shows a negative outlier, with an increase of 10 percentage points in cases where none of the ground-truth papers could be retrieved. Together with questions from the Interdisciplinary category it has the lowest recall on ground truth papers across the dataset. Mathematics and Interdisciplinary questions show the highest partial retrieval, with a slight increase

in cases where one of the two ground truth papers could be retrieved, in comparison with the overall distribution. The strongest deviation is observed for the Other category, however due to low sample size, it represents a statistical outlier rather than an indication of superior retrieval performance.

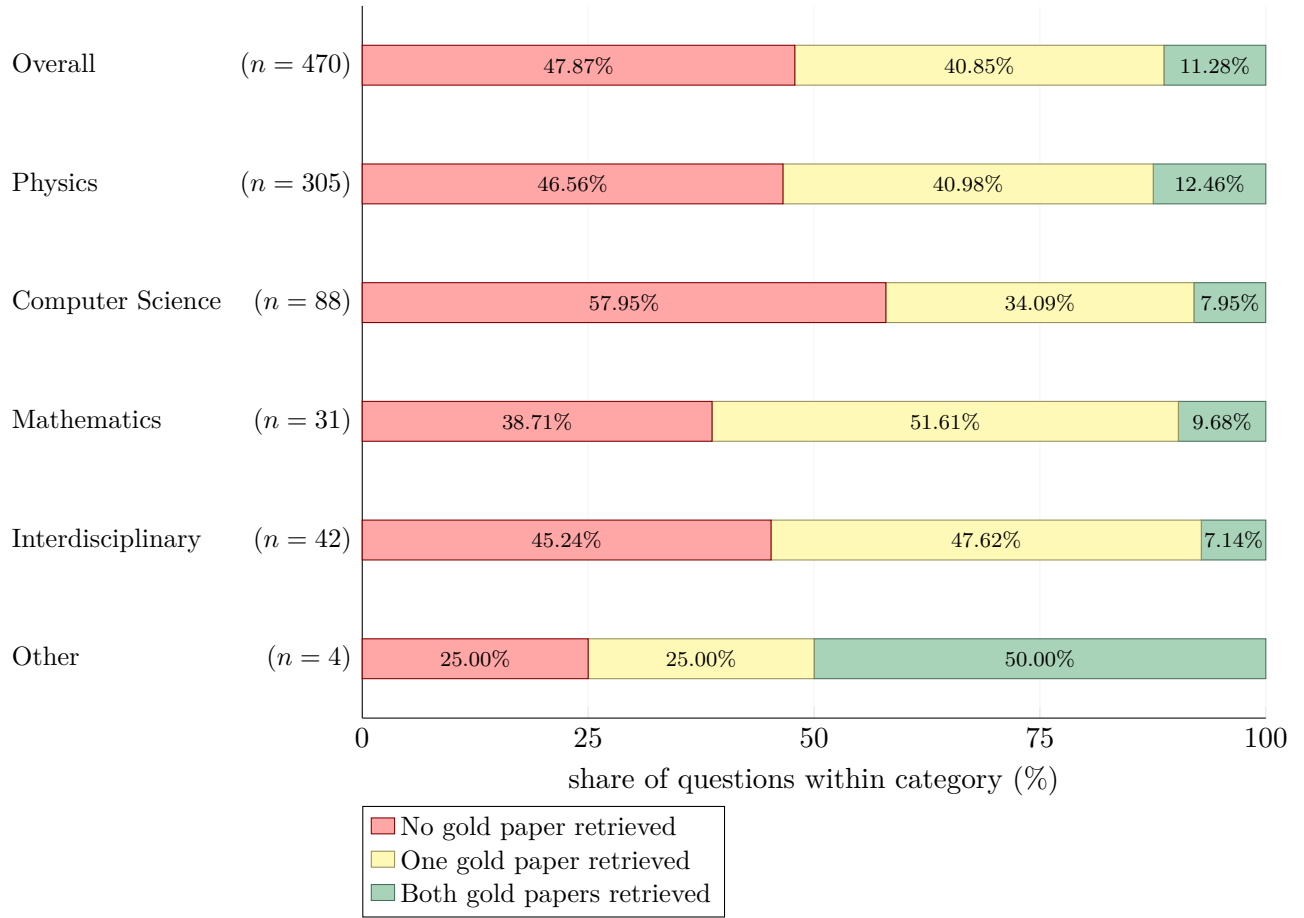


Figure 4.7.: Retrieved gold-ground-truth documents by category

4.3. System evaluation

Bibliography

- [1] Tobias Schreieder, Tim Schopf, and Michael Färber. *Attribution, Citation, and Quotation: A Survey of Evidence-based Text Generation with Large Language Models*. 2025. arXiv: 2508.15396 [cs.CL]. URL: <https://arxiv.org/abs/2508.15396>.
- [2] Jie Huang and Kevin Chang. “Citation: A Key to Building Responsible and Accountable Large Language Models”. In: *Findings of the Association for Computational Linguistics: NAACL 2024*. Ed. by Kevin Duh, Helena Gomez, and Steven Bethard. Mexico City, Mexico: Association for Computational Linguistics, June 2024, pp. 464–473. DOI: 10.18653/v1/2024.findings-naacl.31. URL: <https://aclanthology.org/2024.findings-naacl.31/>.
- [3] Ines Besrouer et al. “SQuAI: Scientific Question-Answering with Multi-Agent Retrieval-Augmented Generation”. In: *Proceedings of the 34th ACM International Conference on Information and Knowledge Management*. CIKM ’25. ACM, Nov. 2025, pp. 6603–6608. DOI: 10.1145/3746252.3761471. URL: <http://dx.doi.org/10.1145/3746252.3761471>.
- [4] Tarek Saier, Johan Krause, and Michael Färber. “unarXive 2022: All arXiv Publications Pre-Processed for NLP, Including Structured Full-Text and Citation Network”. In: *2023 ACM/IEEE Joint Conference on Digital Libraries (JCDL)*. IEEE, 2023, pp. 66–70. DOI: 10.1109/jcdl57899.2023.00020. URL: <http://dx.doi.org/10.1109/JCDL57899.2023.00020>.
- [5] Pradeep Dasigi et al. *A Dataset of Information-Seeking Questions and Answers Anchored in Research Papers*. 2021. arXiv: 2105.03011 [cs.CL]. URL: <https://arxiv.org/abs/2105.03011>.
- [6] Tianyu Gao et al. *Enabling Large Language Models to Generate Text with Citations*. 2023. arXiv: 2305.14627 [cs.CL]. URL: <https://arxiv.org/abs/2305.14627>.
- [7] Shraman Pramanick, Rama Chellappa, and Subhashini Venugopalan. *SPIQA: A Dataset for Multimodal Question Answering on Scientific Papers*. 2025. arXiv: 2407.09413 [cs.CL]. URL: <https://arxiv.org/abs/2407.09413>.
- [8] Yoonjoo Lee et al. “QASA: Advanced Question Answering on Scientific Articles”. In: *Proceedings of the 40th International Conference on Machine Learning*. Ed. by Andreas Krause et al. Vol. 202. Proceedings of Machine Learning Research. PMLR, July 2023, pp. 19036–19052. URL: <https://proceedings.mlr.press/v202/lee23n.html>.
- [9] Chaitanya Malaviya et al. *ExpertQA: Expert-Curated Questions and Attributed Answers*. 2024. arXiv: 2309.07852 [cs.CL]. URL: <https://arxiv.org/abs/2309.07852>.
- [10] Shruti Singh, Nandan Sarkar, and Arman Cohan. “SciDQA: A Deep Reading Comprehension Dataset over Scientific Papers”. In: *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Ed. by Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen. Miami, Florida, USA: Association for Computational Linguistics, Nov. 2024, pp. 20908–20923. DOI: 10.18653/v1/2024.emnlp-main.1163. URL: <https://aclanthology.org/2024.emnlp-main.1163/>.

- [11] Tim Baumgärtner, Ted Briscoe, and Iryna Gurevych. *PeerQA: A Scientific Question Answering Dataset from Peer Reviews*. 2025. arXiv: 2502.13668 [cs.CL]. URL: <https://arxiv.org/abs/2502.13668>.
- [12] Yash Saxena et al. *Generation-Time vs. Post-hoc Citation: A Holistic Evaluation of LLM Attribution*. 2025. arXiv: 2509.21557 [cs.CL]. URL: <https://arxiv.org/abs/2509.21557>.
- [13] Gorjan Radevski et al. *On Synthesizing Data for Context Attribution in Question Answering*. 2025. arXiv: 2504.05317 [cs.IR]. URL: <https://arxiv.org/abs/2504.05317>.
- [14] Ori Ernst et al. “Where Did That Come From? Sentence-Level Error-Tolerant Attribution”. In: *Findings of the Association for Computational Linguistics: EMNLP 2025*. Ed. by Christos Christodoulopoulos et al. Suzhou, China: Association for Computational Linguistics, Nov. 2025, pp. 6400–6417. ISBN: 979-8-89176-335-7. DOI: 10.18653/v1/2025.findings-emnlp.339. URL: <https://aclanthology.org/2025.findings-emnlp.339/>.
- [15] Ansel MacLaughlin et al. *Context-Based Quotation Recommendation*. 2020. arXiv: 2005.08319 [cs.CL]. URL: <https://arxiv.org/abs/2005.08319>.
- [16] Sriram Balasubramanian et al. *Decomposition-Enhanced Training for Post-Hoc Attributions In Language Models*. 2025. arXiv: 2510.25766 [cs.CL]. URL: <https://arxiv.org/abs/2510.25766>.
- [17] Masana Khanyi, Milandre Lill, and Marcel Dunaiski. “Citation Context Extraction Using a Transformer-Based Model with a Question-Answering Task”. In: July 2025, pp. 293–306. ISBN: 978-3-031-96261-5. DOI: 10.1007/978-3-031-96262-2_19.
- [18] Eran Hirsch et al. *LAQuer: Localized Attribution Queries in Content-grounded Generation*. 2025. arXiv: 2506.01187 [cs.CL]. URL: <https://arxiv.org/abs/2506.01187>.
- [19] Anirudh Phukan et al. *Peering into the Mind of Language Models: An Approach for Attribution in Contextual Question Answering*. 2024. arXiv: 2405.17980 [cs.CL]. URL: <https://arxiv.org/abs/2405.17980>.
- [20] Samyadeep Basu et al. *On Mechanistic Circuits for Extractive Question-Answering*. 2025. arXiv: 2502.08059 [cs.CL]. URL: <https://arxiv.org/abs/2502.08059>.
- [21] Michele Bevilacqua et al. *Autoregressive Search Engines: Generating Substrings as Document Identifiers*. 2022. arXiv: 2204.10628 [cs.CL]. URL: <https://arxiv.org/abs/2204.10628>.
- [22] Jiangui Chen et al. “GERE: Generative Evidence Retrieval for Fact Verification”. In: *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*. SIGIR ’22. ACM, July 2022, pp. 2184–2189. DOI: 10.1145/3477495.3531827. URL: <http://dx.doi.org/10.1145/3477495.3531827>.
- [23] Liwen Zheng et al. “MRR-FV: Unlocking Complex Fact Verification with Multi-Hop Retrieval and Reasoning”. In: *Proceedings of the AAAI Conference on Artificial Intelligence* 39.24 (Apr. 2025), pp. 26066–26074. DOI: 10.1609/aaai.v39i24.34802. URL: <https://ojs.aaai.org/index.php/AAAI/article/view/34802>.
- [24] Kishore Papineni et al. “Bleu: a Method for Automatic Evaluation of Machine Translation”. In: *Annual Meeting of the Association for Computational Linguistics*. 2002. URL: <https://api.semanticscholar.org/CorpusID:11080756>.

- [25] Chin-Yew Lin. “ROUGE: A Package for Automatic Evaluation of Summaries”. In: *Text Summarization Branches Out*. Barcelona, Spain: Association for Computational Linguistics, July 2004, pp. 74–81. URL: <https://aclanthology.org/W04-1013/>.
- [26] Tianyi Zhang et al. *BERTScore: Evaluating Text Generation with BERT*. 2020. arXiv: 1904.09675 [cs.CL]. URL: <https://arxiv.org/abs/1904.09675>.
- [27] Wei Zhao et al. *MoverScore: Text Generation Evaluating with Contextualized Embeddings and Earth Mover Distance*. 2019. arXiv: 1909.02622 [cs.CL]. URL: <https://arxiv.org/abs/1909.02622>.
- [28] Weizhe Yuan, Graham Neubig, and Pengfei Liu. *BARTScore: Evaluating Generated Text as Text Generation*. 2021. arXiv: 2106.11520 [cs.CL]. URL: <https://arxiv.org/abs/2106.11520>.
- [29] Hannah Rashkin et al. *Measuring Attribution in Natural Language Generation Models*. 2022. arXiv: 2112.12870 [cs.CL]. URL: <https://arxiv.org/abs/2112.12870>.
- [30] Or Honovich et al. *TRUE: Re-evaluating Factual Consistency Evaluation*. 2022. arXiv: 2204.04991 [cs.CL]. URL: <https://arxiv.org/abs/2204.04991>.
- [31] Wojciech Kryściński et al. *Evaluating the Factual Consistency of Abstractive Text Summarization*. 2019. arXiv: 1910.12840 [cs.CL]. URL: <https://arxiv.org/abs/1910.12840>.
- [32] Philippe Laban et al. *SummaC: Re-Visiting NLI-based Models for Inconsistency Detection in Summarization*. 2021. arXiv: 2111.09525 [cs.CL]. URL: <https://arxiv.org/abs/2111.09525>.
- [33] Or Honovich et al. “Q²: Evaluating Factual Consistency in Knowledge-Grounded Dialogues via Question Generation and Question Answering”. In: *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*. Ed. by Marie-Francine Moens et al. Online and Punta Cana, Dominican Republic: Association for Computational Linguistics, Nov. 2021, pp. 7856–7870. DOI: 10.18653/v1/2021.emnlp-main.619. URL: <https://aclanthology.org/2021.emnlp-main.619/>.
- [34] Jinlan Fu et al. *GPTScore: Evaluate as You Desire*. 2023. arXiv: 2302.04166 [cs.CL]. URL: <https://arxiv.org/abs/2302.04166>.
- [35] Yang Liu et al. *G-Eval: NLG Evaluation using GPT-4 with Better Human Alignment*. 2023. arXiv: 2303.16634 [cs.CL]. URL: <https://arxiv.org/abs/2303.16634>.
- [36] Jason Wei et al. *Chain-of-Thought Prompting Elicits Reasoning in Large Language Models*. 2023. arXiv: 2201.11903 [cs.CL]. URL: <https://arxiv.org/abs/2201.11903>.
- [37] Lianmin Zheng et al. *Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena*. 2023. arXiv: 2306.05685 [cs.CL]. URL: <https://arxiv.org/abs/2306.05685>.
- [38] Sewon Min et al. *FActScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation*. 2023. arXiv: 2305.14251 [cs.CL]. URL: <https://arxiv.org/abs/2305.14251>.
- [39] Xiang Yue et al. *Automatic Evaluation of Attribution by Large Language Models*. 2023. arXiv: 2305.06311 [cs.CL]. URL: <https://arxiv.org/abs/2305.06311>.

- [40] Yumo Xu et al. “CiteEval: Principle-Driven Citation Evaluation for Source Attribution”. In: *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Ed. by Wanxiang Che et al. Vienna, Austria: Association for Computational Linguistics, July 2025, pp. 32759–32778. ISBN: 979-8-89176-251-0. DOI: 10.18653/v1/2025.acl-long.1574. URL: <https://aclanthology.org/2025.acl-long.1574/>.
- [41] Shahul Es et al. *Ragas: Automated Evaluation of Retrieval Augmented Generation*. 2025. arXiv: 2309.15217 [cs.CL]. URL: <https://arxiv.org/abs/2309.15217>.
- [42] Yifei Li et al. *AttributionBench: How Hard is Automatic Attribution Evaluation?* 2024. arXiv: 2402.15089 [cs.CL]. URL: <https://arxiv.org/abs/2402.15089>.
- [43] Jon Saad-Falcon et al. *ARES: An Automated Evaluation Framework for Retrieval-Augmented Generation Systems*. 2024. arXiv: 2311.09476 [cs.CL]. URL: <https://arxiv.org/abs/2311.09476>.
- [44] Arman Cohan et al. “SPECTER: Document-level Representation Learning using Citation-informed Transformers”. In: *ArXiv abs/2004.07180* (2020). URL: <https://api.semanticscholar.org/CorpusID:215768677>.
- [45] URL: <https://api.semanticscholar.org/api-docs/>.
- [46] Norman Paulsen. *Context Is What You Need: The Maximum Effective Context Window for Real World Limits of LLMs*. Sept. 2025. DOI: 10.48550/arXiv.2509.21361.
- [47] URL: <https://huggingface.co/moonshotai/Kimi-K2.6>.
- [48] Cheng-Ping Hsieh et al. *RULER: What’s the Real Context Size of Your Long-Context Language Models?* 2024. arXiv: 2404.06654 [cs.CL]. URL: <https://arxiv.org/abs/2404.06654>.
- [49] Weiwei Wang, Jiyong Min, and Weijie Zou. *Intelligence Degradation in Long-Context LLMs: Critical Threshold Determination via Natural Length Distribution Analysis*. 2026. arXiv: 2601.15300 [cs.CL]. URL: <https://arxiv.org/abs/2601.15300>.
- [50] Jonathan Roberts, Kai Han, and Samuel Albanie. *How Long Is a Piece of String? A Brief Empirical Analysis of Tokenizers*. 2026. arXiv: 2601.11518 [cs.CL]. URL: <https://arxiv.org/abs/2601.11518>.
- [51] URL: [openai/gpt-oss-120b](https://openai.com/gpt-oss-120b).
- [52] Harsh Trivedi et al. “Is Multihop QA in DiRe Condition? Measuring and Reducing Disconnected Reasoning”. In: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Ed. by Bonnie Webber et al. Online: Association for Computational Linguistics, Nov. 2020, pp. 8846–8863. DOI: 10.18653/v1/2020.emnlp-main.712. URL: <https://aclanthology.org/2020.emnlp-main.712/>.

A. Appendix

```

1  You are generating a scientific 2-hop question-answer-evidence (Q-A) Tuple.
2  You are given 5 candidate scientific papers in the following format:
3  [
4      {
5          "ArXiv": string,
6          "text": string,
7      }
8  ]
9
10 Your task is to select the best pair of papers and generate ONE question that
    requires connected reasoning across exactly TWO papers.
11
12 PAPER SELECTION
13 First, select the best pair of papers.
14 The selected pair must satisfy:
15
16 -Paper A must provide more than a named entity. It must provide an operational
    meaning, condition, mechanism, definition, directionality, limitation, assumption
    , guarantee, interpretation, failure mode, objective, result or similar that is
    needed to interpret Paper B.
17 -Paper B must provide a concrete use case, measurement, experiment, result,
    comparison, design choice, observation, conclusion or similar whose meaning
    depends on the information from Paper A.
18 -Paper A must provide an intermediate proposition about an entity, method, dataset,
    variable, result, or concept, not merely the name of that entity.
19 -Paper B must require applying that proposition, not merely recognizing that the same
    entity is used again.
20
21 -A question is invalid if Paper A is only used to identify the name of a method,
    metric, model, dataset or similar that Paper B already mentions, so you cannot
    just describe the selected entity loosely and ask for identification or naming of
    it.
22 The final answer must not be only the name of a method, metric, dataset, model, or
    concept. It must be an explanation, conclusion, comparison, interpretation, or
    derived statement that depends on combining both papers.
23 -The final answer requires combining both papers
24 -Do NOT select papers that are only loosely related or redundant.
25
26 Do not generate questions whose reasoning type is only:
27 -entity linking,
28 -artifact reuse,
29 -citation tracing,
30 -numeric lookup,
31 -table lookup,
32 -independent comparison,

```

```

33 -descriptive clue matching.
34
35 REQUIREMENTS
36 A valid question MUST:
37 -require exactly two reasoning steps (2-hop)
38 -use exactly two supporting papers
39 -require sequential reasoning:
40     -Step 1 must derive a proposition from Paper A, not merely identify a named
        entity, metric, model, dataset or benchmark. Instead it should derive an
        interpretable proposition from Paper A such as a mechanism, condition,
        limitation, guarantee, assumption, theoretical relation, or failure mode.
41     -Step 2 must apply the proposition from Step 1 to interpret a concrete result,
        design choice, experiment, observation, or conclusion from Paper B.
42     -Step 2 must not be solvable by reading Paper B alone, even if Paper B mentions
        the same method, metric, model, dataset, benchmark, architecture, or loss
        function. Thus Paper A must be strictly required to answer the question.
43     -If the answer can be produced by first identifying an artifact from Paper A and
        then looking up how Paper B uses that artifact, the question is invalid.
44 -NOT be answerable from either paper alone
45 -be self-contained and unambiguous
46 -require combining information across both papers
47 -A valid question must require real reasoning rather than entity linking. The answer
        must depend on applying a proposition from Paper A to interpret evidence from
        Paper B, not merely identifying a named artifact or matching similar wording
        across papers.
48 -only use information that is present in the papers, do not invent hypothetical
        connections or applications. Only use existing connections.
49 -The question may mention the concept from Paper A, but it must not state the full
        intermediate conclusion from Paper A as an already-given premise. The respondent
        should still need Paper A to derive or justify Step 1.
50 -question cannot contain explicit references to the papers or its content such as "in
        this paper", "the proposed methods" or similar
51 -The question may use necessary technical terminology from the papers, but it should
        not copy distinctive phrasing from either paper unless the phrase is a standard
        technical term. Rephrase the evidence into a reasoning-focused question.
52
53
54 AVOID
55 Do NOT generate:
56 -multi-part questions (e.g., "What is X and what is Y?")
57 -vague or generic questions (e.g., "implications", "advantages")
58 -literature summary questions
59 -questions answerable from a single paper
60 -questions where papers are only topically related but not logically connected
61 -questions where Paper A only introduces an artifact and Paper B merely reuses,
        applies, evaluates, compares against, cites, or optimizes that artifact or where
        the answer is just the artifact itself or a verbatim description of it

```

```

62 -questions where the two steps are independent facts that can be answered separately
    and simply concatenated
63 -questions where Step 1 can be solved mainly by matching a distinctive phrase in the
    question to nearly identical wording in Paper A
64
65 -Do NOT use external scientific knowledge, commonsense assumptions, or unstated
    domain knowledge, only use the information that is present in the papers
66
67 Invalid:
68 Paper A defines method X. Paper B uses method X.
69 Paper A introduces dataset X. Paper B evaluates on X.
70 Paper A introduces metric X. Paper B reports metric X.
71 Paper A describes architecture X. Paper B applies X.
72
73 OUTPUT FORMAT
74 {
75     "usedPapers" : [
76         {
77             "arXiv": <paperId>,
78             "role": <bridgeEvidence / bridgeAnswer depending on the papers role>
79         },
80         {
81             "arXiv": <paperId>,
82             "role": <bridgeEvidence/ bridgeAnswer depending on the papers role>
83         }
84     ],
85     "rejectedPapers": [<paperId1>, <paperId2>, <paperId3>],
86     "reasoning": {
87         "bridgeProposition": <the interpretable proposition derived from Paper A,
            such as a mechanism, condition, limitation, guarantee, theoretical
            relation, or failure mode; this must not be merely a named artifact>,
88         "paperB0bservation": <the concrete result, design choice, experiment,
            observation, or conclusion from Paper B that requires interpretation>,
89         "appliedInference": <explain the connection>,
90         "step1": <Explain your reasoning for step1 on paper A>,
91         "step2": <use step1 reasoning to derive or support the final answer>,
92         "connectionExplanation": <explain why Step 2 cannot be solved without Step 1,
            and why this is not entity linking, artifact reuse, table lookup, numeric
            lookup, or lexical matching>
93     },
94     "questionDraft" : "<one clear, self-contained question requiring 2-hop reasoning,
        this serves as a draft for you final question. You can but don't have to
        explicitly reference the papers here to help formulate a better question.>",
95     "question" : <decontextualize the questionDraft, so that it contains no explicit
        reference to the papers or external figures. Every reference to a paper
        should instead directly name the concept, phenomenon, method or similar that
        you are referring to, briefly describing it if needed. The question cannot

```

```

    contain explicit references to the papers or its content such as "in this
    paper", "the proposed methods" or similar.>",
96   "answerWithPaperReferences" : <long-form paragraph integrating Paper A and Paper
    B with citations like [Paper A], [Paper B]>,
97   "answerWithoutPaperReferences": <long-form paragraph that answers the question,
    without referencing the papers, but directly incorporates their information,
    so that it is standalone understandable>,
98   "isNotSingleHop": <explain why neither Paper A nor Paper B alone can answer the
    question, why Step 2 requires applying the proposition from Step 1, and why
    the question is not merely artifact reuse, entity linking, or lexical
    matching>
99 }
100
101 If no valid question can be generated, answer only with the exact string "NO_QUESTION
    _GENERATABLE".
102 If a valid question can be generated, answer only with the valid JSON object. You are
    not allowed to deviate from this schema. Do not add any preceding information
    like ``json. Only Answer with the valid json.
103
104 Paper Texts:

```

Listing A.1: Question–Answer generation prompt

```

1  You are evaluating a scientific question-answer (Q-A) example. You are given a
    generated question, an answer and two paper texts. Your goal is to judge how well
    this generated question fullfills a criterion.
2
3  The criterion is:
4  Multi-hop Validity: Does answering the question require combining the information
    from both papers?
5
6  1. Decompose the question into the necessary claims, that need to be answered in
    order to answer the whole question.
7  2. For each required claim, identify which papers answers it:
8     -already stated in the question,
9     -answerable only by Paper 1,
10    -answerable only by Paper 2,
11    -answerable by both papers,
12    -unsupported or inferred.
13  3. Test whether either paper alone is sufficient:
14     Could Paper 1 alone answer the complete question?
15     Could Paper 2 alone answer the complete question?
16  4. Test whether question leakage makes one paper unnecessary:
17     Does the question already state the main fact, mechanism, definition, objective,
        result, or condition that one paper is supposed to contribute?
18     Could the other paper plus the information already stated in the question answer
        the main question?
19

```

```

20 5. Check whether both papers provide non-redundant required information:
21   If one paper only identifies, introduces, defines, or names an artifact that the
      other paper already properly introduces, uses, mentions, evaluates, optimizes,
      cites, or compares against, this is not enough for GOOD. However, a good
      judgement here can be possible, if the other paper adds a non-trivial
      proposition about that artifact, required for interpreting, analyzing,
      applying or justifying or similar.
22 6. Distinguish the main answer from optional elaboration.
23   Ask whether both papers are required for the central answer, not just for a richer
      or more detailed answer. If one paper answers the main question and the other
      only adds nuance, background, or a fuller mechanism, assign BORDERLINE rather
      than GOOD.
24
25 Finally, based on the previous steps, give a final judgement whether the question
      truly is multihop- valid.
26 JUDGEMENT OPTIONS:
27 -GOOD: Both papers are strictly required; neither paper alone provides enough
      information to answer the complete question.
28 -BORDERLINE: Both papers contribute, but one paper may be sufficient to answer the
      main question, or the second paper mainly adds background/context.
29 -BAD: Only one paper is sufficient; the question is effectively single-hop.
30 Important: If one paper only identifies, introduces, defines, or names an artifact
      that the other paper already uses, mentions, evaluates, optimizes, cites, or
      compares against, this is not enough for GOOD. The answer must require combining
      non-redundant evidence from both papers.
31
32 INPUT:
33 Question: {question}
34 Answer: {answer}
35 bridgeEvidencePaperText: {paper1Text}
36 bridgeAnswerPaperText: {paper2Text}
37
38 OUTPUT_FORMAT
39 {{
40   "thinking": {{
41     "claims": [
42       {{
43         "claim": <The first claim you identified>,
44         "sourceSupportedBy": <question | paper1 | paper2 | both | unsupported_or_inferred>
45       }},
46       ...
47     ],
48     "paper1AloneSufficient": {{
49       "judgement": <true | false>,
50       "explanation": <explain why or why not paper 1 alone is sufficient for answering the
      question>
51     }},
52     "paper2AloneSufficient": {{

```



```

53  "judgement": <true | false>,
54  "explanation": <explain why or why not paper 2 alone is sufficient for answering the
      question>
55  }},
56  "paper1PlusQuestionSufficient": {{
57  "judgement": <true | false>,
58  "explanation": <explain why or why not paper 1 together with the information in the
      question is sufficient for answering the question>
59  }},
60  "paper2PlusQuestionSufficient": {{
61  "judgement": <true | false>,
62  "explanation": <explain why or why not paper 2 together with the information in the
      question is sufficient for answering the question>
63  }},
64  "questionLeakage": {{
65  "present": <true | false>,
66  "explanation": <explain why or why no question leakage is present>
67  }},
68  "redundancyOrArtifactOnly": {{
69  "present": <true | false>,
70  "explanation": <explain why or why not >
71  }},
72  "mainVsOptionalElaboration": {{
73  "present": <true | false>,
74  "explanation": <explain why or why not >
75  }}
76  }},
77  "judgement": "<insert the final judgement, considering all previous evaluations here.
      Are both papers really required. Answer with: GOOD | BORDERLINE | BAD>",
78  "explanation": "<thoroughly describe how good the question fulfills the multihop
      validity criteria and justify your judgement based on previous evaluations>"
79  }}
80
81  Do not deviate from this schema. Do not add any preceding information like ‘‘json.
      Only Answer with the valid json

```

Listing A.2: Multihop-Validity Prompt

```

1      You are evaluating a scientific question-answer (Q-A) example. You are given a
      generated question, an answer, reasoning as well as two paper texts. Your
      goal is to judge how well this generated question fullfills a criterion.
2
3      The criterion is:
4      Dependency Strength: Does Step 2 depend on the result of Step 1?
5
6      1. Identify the Step 1 proposition.
7      Ask: What exact mechanism, condition, definition, limitation, assumption,
      objective, result, or interpretation must be derived from Paper 1?

```

```

8
9 2. Check whether Step 1 is non-trivial.
10 Step 1 should be more than identifying a method, artifact, dataset, metric, or
    named entity.
11
12 3. Check whether Step 1 is already leaked by the question.
13 If the question already states the Step 1 proposition, then Step 2 does not
    strongly depend on deriving it from Paper 1.
14
15 4. Identify the Step 2 claim.
16 Ask: What exact result, application, observation, comparison, or finding comes
    from Paper 2?
17
18 5. Test strict dependency.
19 Ask: Could Step 2 be answered or interpreted without applying the Step 1
    proposition?
20 If yes, dependency is weak.
21
22 6. Check for entity-linking.
23 If Step 1 only identifies an artifact/method and Step 2 only tracks that artifact/
    method in another paper, assign BAD or BORDERLINE, not GOOD.
24
25 Finally, based on the previous steps, give a final judgement whether the question
    consists of 2 consecutive steps, where step 2 is only answerable by applying the
    results of step 1.
26
27 JUDGEMENT OPTIONS:
28 -GOOD: Step 2 strictly requires applying a proposition derived from Step 1. Step 1
    must provide more than a named entity; it must provide a mechanism, condition,
    definition, limitation, assumption, guarantee, interpretation, objective, result,
    or similar.
29 -BORDERLINE: Step 2 is related to Step 1, but the dependency is weak, mostly entity-
    linking, or one step can mostly be solved without the other.
30 -BAD: Steps are independent, disconnected, or Step 1 only identifies a name/entity
    used in Step 2.
31
32 INPUT:
33 Question: {question}
34 Answer: {answer}
35 reasoning : {reasoning}
36 bridgeEvidencePaperText: {paper1Text}
37 bridgeAnswerPaperText: {paper2Text}
38
39 OUTPUT_FORMAT
40 {{
41 "thinking": {{
42 "step1Claim": <Describe the mechanism you extracted from step1>,
43 "step1NotTrivial": {{

```

```

44  "judgement": <true | false>,
45  "explanation": <Describe why or why not step 1 is trivial>
46  }},
47  "step1LeakedByQuestion": {{
48  "judgement": <true | false>,
49  "explanation": <Describe why or why not step 1 is leaked by the question>
50  }},
51  "step2Claim": <Describe the mechanism you extracted from step2>,
52  "strictDependency": {{
53  "judgement": <true | false>,
54  "explanation": <Describe why or why not step2 is stricly dependant on step1>
55  }},
56  "entityLinking": {{
57  "judgement": <true | false>,
58  "explanation": <Describe why or why not the question is just entity linking>
59  }},
60  }},
61  "judgement": "<insert the final judgement, considering all previous evaluations here.
        Are both papers really required. Answer with: GOOD | BORDERLINE | BAD>",
62  "explanation": "<thoroughly describe how good the question fulfills the multihop
        validity criteria and justify jour judgement based on previous evaluations>"
63  }}
64
65  Do not deviate from this schema. Do not add any preciding information like ‘‘‘json.
        Only Answer with the valid json

```

Listing A.3: Dependency Strength Prompt

```

1      You are evaluating a scientific question-answer (Q-A) example. You are given a
      generated question, an answer and two paper texts. Your goal is to judge how
      well this generated question fullfills a criterion.
2
3      The criterion is: Non-Decomposability: Is the question NOT decomposable into
      independent sub-questions?
4
5      1. Ask what the answer for the question must explain, compare, justify, or conclude.
6      2. Try to decompose the question into independent sub-questions.
7      For example:
8      What does Paper 1 say about X?
9      What does Paper 2 say about Y?
10     What result does Paper 2 report?
11     What mechanism does Paper 1 describe?
12     3. Check whether each sub-question can be answered independently.
13     Ask whether the Paper 1 part and the Paper 2 part can each be solved without using
        the other.
14     4. Check whether the final answer is just concatenation or comparison.

```

```

15     If the answer can be produced by separately retrieving facts from each paper and
16       then simply joining them with "therefore," "in contrast," or "because," the
17       question is decomposable.
18
19     5. Check whether solving one part changes, constrains, or enables solving the other.
20       GOOD requires that one sub-answer materially affects how the other sub-answer is
21       interpreted or answered.
22
23     6. Check for artificial integration.
24       If the question appears integrated only because it mentions both papers, both
25       methods, or both results, but the reasoning can still be split into
26       independent lookups, assign BAD or BORDERLINE.
27
28     7. Check whether the explanatory bridge is necessary.
29       If the question asks for "why" or "how," determine whether the bridge requires
30       genuine synthesis or whether it is just a plausible comparison between
31       independently retrieved facts.
32
33       Decide whether the question is truly non-decomposable.
34
35       If the answer requires integrated reasoning where one part constrains or enables
36       the other, assign GOOD.
37
38       If the answer needs some interpretation but the sub-answers are mostly independent
39       , assign BORDERLINE.
40
41       If the answer is mainly independent lookups plus concatenation/comparison, assign
42       BAD.
43
44
45     Finally, based on the previous steps, give a final judgement whether the question is
46     decomposable into independent sub-questions.
47
48     JUDGEMENT OPTIONS
49
50     -GOOD: The question cannot be split into independent single-hop questions; solving
51     one part changes, constrains, or enables solving the other.
52
53     -BORDERLINE: The question is partially decomposable, but the sub-answers still need
54     some linking or interpretation.
55
56     -BAD: The question clearly decomposes into independent sub-questions whose answers
57     can simply be concatenated.
58
59
60     INPUT:
61
62     Question: {question}
63     Answer: {answer}
64     bridgeEvidencePaperText: {paper1Text}
65     bridgeAnswerPaperText: {paper2Text}
66
67
68     OUTPUT_FORMAT
69
70     {{
71     "thinking": {{
72     "mainQuestion": <Describe what the question must answer, explain, discuss, judge or
73       similar>,
74     "subquestions": [
75     {{
76     "question": <The first question that was the result of your question decomposition >

```

```

46  "independentlyAnswerableBy": <give a verdict, wether the question can be
    independently solved by using either of the 3 options PAPER1 | PAPER2 | BOTH_
    PAPERS_TOGETHER>
47  }},
48  ...
49  ],
50  "concatenationOrComparison": {{
51      "judgement": <true | false>,
52      "explanation": "<explain whether the final answer can be produced by
    concatenating or simply comparing separately retrieved facts>"
53  }},
54  "onePartConstrainsOther": {{
55      "judgement": <true | false>,
56      "explanation": "<explain whether solving one part changes, constrains, or
    enables solving the other>"
57  }},
58  "artificialIntegration": {{
59      "judgement": <true | false>,
60      "explanation": "<explain whether the question only appears integrated
    because it mentions both papers, methods, entities, or results>"
61  }},
62  "necessaryExplanatoryBridge": {{
63      "judgement": <true | false>,
64      "explanation": "<explain whether the explanatory bridge requires genuine
    synthesis or is just a plausible comparison>"
65  }}
66  }},
67  "judgement": "<insert the final judgement, considering all previous evaluations here.
    Are both papers really required. Answer with: GOOD | BORDERLINE | BAD>",
68  "explanation": "<thoroughly describe how good the question fulfills the multihop
    validity criteria and justify your judgement based on previous evaluations>"
69  }}
70
71  Do not deviate from this schema. Do not add any preciding information like ‘‘json.
    Only Answer with the valid json

```

Listing A.4: Non-Decomposability Prompt

```

1      You are evaluating a scientific question-answer (Q-A) example. You are given a
    generated question, an answer and two paper texts. Your goal is to judge how
    well this generated question fullfills a criterion.
2
3      The criterion is: Evidence Distribution: Are both papers required and non-redundant?
4
5      Important:
6
7      - The answer is a candidate answer, not evidence.
8      - Use the answer only as a clue to the intended response.

```

9 - Judge what evidence is actually required to answer the question.
10 - Verify all claims directly against the two paper texts.
11 - Be strict: assign GOOD only if both papers provide distinct, necessary, non-
redundant evidence.

12
13 Judging procedure:
14

15 1. Identify the evidence needed to answer the question.
16 Break the answer into evidence pieces, not just topics or method names.
17

18 2. List exactly what Paper 1 contributes.
19 Ask whether Paper 1 provides a necessary intermediate proposition, mechanism,
condition, definition, limitation, assumption, objective, result, or
interpretation.
20 Ask which of the evidence from 1. it contributes.
21

22 3. List exactly what Paper 2 contributes.
23 Ask whether Paper 2 provides necessary evidence whose meaning depends on applying
the Paper 1 contribution.
24 Ask which of the evidence from 1. it contributes.
25

26 4. Check whether either paper is dominant.
27 If one paper contains the task, result, comparison, method description, and enough
context to answer the main question, while the other only adds support or
detail, Evidence Distribution is not GOOD.
28

29 5. Check for redundancy or overlap.
30 Ask whether both papers provide the same or overlapping information regarding the
evidence needed for answering the question, or whether one paper repeats/
elaborates what the other paper or the question already states.
31

32 6. Check for thin or artifact-only evidence roles.
33 Ask whether one paper's contribution is only naming, defining, or describing an
artifact/method while the other paper provides the substantive task-specific
evidence. If so, the evidence distribution is weak because one paper has only
a background/detail role.
34

35 Finally, based on the previous steps, give a final judgement whether the question
targets information distributed above both papers.

36 Judgement options:
37 -GOOD: Each paper contributes distinct, necessary information. One paper provides an
intermediate proposition, and the other provides evidence whose meaning depends
on applying that proposition.
38 -BORDERLINE: Both papers contribute, but there is overlap, redundancy, or one paper
is dominant while the other only adds support.
39 -BAD: One paper is sufficient; the other is redundant, only background, or only
provides the name/source of an artifact.
40

```

41  Input:
42  Question: {question}
43  Answer: {answer}
44  paper1Text: {paper1Text}
45  paper2Text: {paper2Text}
46
47  OUTPUT_FORMAT
48  {{
49  "thinking": {{
50  "evidenceNeeded": [
51  <List evidence that is needed to answer the question>
52  ],
53  "evidencePaper1": [
54  <List evidence that paper 1 contributes>
55
56      ],
57      "evidencePaper2": [
58          <List evidence that paper 2 contributes>
59      ],
60      "paperDominance": {{
61          "dominantPaper": <answer with either PAPER1 | PAPER2 | NONE depending on
              if a paper is dominant or not>,
62          "explanation": <explain why or why not either paper is dominant>
63      }},
64      "informationRedundancy": {{
65          "judgement": <true | false>,
66          "explanation": <explain why or why not the information from both papers is
              redundant>
67      }},
68      "artifactOnlyEvidence": {{
69          "judgement": <true | false>,
70          "explanation": <explain why or why not one paper's contribution is
              artifact only evidence>
71      }}
72  }},
73  "judgement": "<insert the final judgement, considering all previous evaluations
              here. Does the question target evidence distributed across both papers?
              Answer with: GOOD | BORDERLINE | BAD>",
74  "explanation": "<thoroughly describe how good the question fulfills the evidence
              distribution criteria and justify your judgement based on previous
              evaluations>"
75  }}
76  Do not deviate from this schema. Do not add any preceding information like ``json.
    Only Answer with the valid json

```

Listing A.5: Evidence Distribution Prompt

1 You are evaluating a scientific question-answer (Q-A) example. You are given a
2 generated question, a candidate answer, and two paper texts. Your goal is to
3 judge how well the candidate answer fulfills this criterion:

4

5 Answerability: Is the answer fully supported by the provided evidence?

6

7 Important:

8 - The answer is a candidate answer, not evidence.

9 - Verify the answer directly against the two paper texts.

10 - Do not use external knowledge.

11 - Be strict: assign GOOD only if all central parts of the answer are fully supported
 by the provided papers.

12 - If the answer contains an explanation such as "because", "therefore", "why, "
 explains", "accounts for" or "makes X better/worse" verify that this explanatory
 link is supported, not just the individual facts

13 Judging procedure:

14

15 1. Break the candidate answer into the minimum claims needed to answer the question.
16 Do not over-penalize irrelevant extra wording unless it affects the answer.

17

18 2. For each answer claim, identify whether it is supported by:

19 - paper1

20 - paper2

21 - both

22 - question_only

23 - unsupported_or_inferred

24

25 3. Check whether every answer claim is directly supported by the paper texts.
26 A good question must have the claims supported by the the paper texts, not by the
 question or unsupported or inferered.

27

28 4. Check whether any explanation or causal bridge is only inferred, instead of backed
 by evidence.

29

30 5. Check whether any answer claim is contradicted by the papers.

31

32 Finally, based on the previous steps, give a final judgement wether the question is
 fully answerable using the papers.

33

34 JUDGEMENT OPTIONS:

35

36 - GOOD: Fully supported by the provided evidence from the papers; no external
 knowledge, speculation, or unsupported inference is needed.

37 - BORDERLINE: Partially supported, but some part of the answer requires mild
 inference, is under-specified, or is not directly grounded.


```

38 - BAD: Not supported, contradicted by the evidence, or requires external knowledge/
    speculation.
39
40 INPUT:
41 Question: {question}
42 Answer: {answer}
43 bridgeEvidencePaperText: {paper1Text}
44 bridgeAnswerPaperText: {paper2Text}
45
46 OUTPUT FORMAT:
47 {{
48   "thinking": {{
49     "answerClaims": [
50       {{
51         "claim": "<The first claim extracted from the answer.>",
52         "supportedBy" <insert by which source this claim is supported, answer with
                    either: paper1 | paper2 | both | question_only | unsupported_or_inferred>,
53         "evidence": <explain why this claim is supported by the source>,
54       }},
55     ...
56   ],
57   "allClaimsSupported": {{
58     "judgement": <true | false>,
59     "explanation": "<explain whether all claims are directly supported by the
                    papers>"
60   }},
61   "unsupportedInferenceOrBridge": {{
62     "present": <true | false>,
63     "explanation": "<explain whether the answer requires inference, an unsupported
                    explanatory bridge, speculation, or external knowledge>"
64   }},
65   "contradiction": {{
66     "present": <true | false>,
67     "explanation": "<explain whether any answer claim is contradicted by the papers
                    >"
68   }}
69   }},
70   "judgement": "<GOOD | BORDERLINE | BAD>",
71   "explanation": "<justify the Answerability judgement using your previous analysis
                    from above>"
72   }}
73
74 Do not deviate from this schema. Do not add any preceding information such as “‘json
    . Only answer with valid JSON.

```

Listing A.6: Answerability Prompt

1	You are evaluating a scientific question-answer (Q-A) example. You are given a generated question, a candidate answer, and two paper texts. Your goal is to judge how well the generated question fulfills this criterion:
2	
3	Decontextualization: Is the question self-contained and unambiguous?
4	
5	Important:
6	
7	- Judge the question itself, not the answer.
8	- Do not penalize technical terminology if the relevant entities, methods, datasets, metrics, or concepts are clearly named or described.
9	- Do not require the question to be easy for a layperson; it only needs to be understandable as a standalone scientific question.
10	- Be strict: assign GOOD only if the question can be understood without seeing the papers.
11	- Do not use this criterion to judge multi-hop validity, evidence distribution, or answerability.
12	
13	Judging procedure:
14	
15	1. Check whether the question contains explicit paper references.
16	Look for phrases such as:
17	- "in this paper"
18	- "the proposed method"
19	- "this approach"
20	- "the authors"
21	- "the study"
22	- "the first paper"
23	- "the second paper"
24	- "the method described above"
25	- "their results"
26	If such references are needed to understand the question, the question is not fully decontextualized.
27	
28	2. Check whether all key entities and concepts are named or clearly described.
29	Ask whether the question clearly identifies the relevant methods, models, datasets, metrics, tasks, phenomena, or domain setting.
30	
31	3. Check for ambiguous pronouns or underspecified references.
32	Look for unclear uses of "it," "this," "that," "the model," "the method," "the metric," "the approach," or similar.
33	
34	4. Check for copied distinctive phrasing that turns the question into a lexical lookup.
35	Technical terminology is allowed, but the question should not merely copy a long distinctive phrase from a paper to make retrieval trivial.
36	

```

37 Finally, based on the previous steps, give a final judgement whether the question is
    self-contained and unambiguous.
38
39 Judgement options:
40
41 - GOOD: The question is understandable on its own, names or describes all required
    entities/concepts, and contains no paper-dependent references.
42 - BORDERLINE: The question is mostly self-contained but has minor ambiguity,
    underspecification, or slightly paper-dependent phrasing.
43 - BAD: The question relies on paper context, explicit paper references, ambiguous
    references, or is not understandable on its own.
44
45 Input:
46 Question: {question}
47 Answer: {answer}
48 paper1Text: {paper1Text}
49 paper2Text: {paper2Text}
50
51 Output format:
52 {{
53   "thinking": {{
54     "explicitPaperReferences": {{
55       "judgement": <true | false>,
56       "examples": [
57         "<quote any explicit paper-dependent reference, or leave empty>"
58       ],
59       "explanation": "<explain whether the question contains explicit references to
    the papers or authors>"
60     }},
61     "keyEntitiesAndConceptsNamed": {{
62       "judgement": <true | false>,
63       "missingOrUnclearItems": [
64         "<state any missing or unclear entity/concept, or leave empty>"
65       ],
66       "explanation": "<explain whether the question names or clearly describes the
    necessary entities and concepts>"
67     }},
68     "ambiguousReferences": {{
69       "judgement": <true | false>,
70       "examples": [
71         "<quote any ambiguous pronoun/reference, or leave empty>"
72       ],
73       "explanation": "<explain whether the question contains ambiguous references>"
74     }},
75     "hiddenPaperContextRequired": {{
76       "present": <true | false>,
77       "explanation": "<explain whether understanding the question requires seeing the
    papers>"

```

```

78     }}
79   }},
80   "judgement": "<GOOD | BORDERLINE | BAD>",
81   "explanation": "<justify the Decontextualization judgement using the diagnostics
      above>"
82   }}
83
84   Do not deviate from this schema. Do not add any preceding information such as ``json
      . Only answer with valid JSON.

```

Listing A.7: decontextualization Prompt

```

1   You are given 2 scientific paper texts as well as a logical connection that exists
    between that papers. Your task is to determine wether that connection is
    explicitelty mentioned in one of the 2 papers, or if is drawn based on their
    content, but not explicitelty mentioned.
2
3   IMPORTANT:
4   -Use only the information from the given papers
5   -Do NOT rely on external knowledge
6   -Be strict: if any critical information is missing, the answer is NOT recoverable
7   -Partial or speculative answers count as NOT answerable
8
9   INPUT:
10  paper1: {paper1}
11  paper2: {paper2}
12  connection: {connection}
13
14  OUTPUT FORMAT
15  {{
16    "explanation" : "<describe your reasoning>",
17    "isPresent": <true/ false depending on if the connection is explicitelty mentioned
      in at least one of the papers, false if not>
18  }}
19  Do not deviate from this schema. Do not add any preceding information like ``json.
    Only Answer with the valid json

```

Listing A.8: DIRE prompt

```

1   You will be given the text of a scientific paper and a claim which was generated
    based on the content of the paper.
2   You can assume that all information in the paper is factually correct. Your job is,
    that for the given claim you should extract the section of the paper which
    provides the
3   best factual basis for justifying the statement. For that you should at least extract
    one sentence and up to 5 sentences which must be consecutively present in the
    paper.

```

```

4   The extraction can only contain sentences that are directly quoted as they are
    written in the document. You are not allowed to make any adjustments whatsoever
    to them.
5   Here is a list of criteria that defines what a good
6   extraction looks like:
7       -The claim is the logical consequence of the facts in the context
8       -The claim and the context cover the same topic
9       -each topic that is present in the claim is also covered in the context
10      -the context does not contradict the claim
11      -the claim includes only information provided in the context or correct
        generalizations or interpretations of that information
12      -the context includes every piece of information needed to justify each
        individual piece of the claim
13      -the context and the claim convey the same meaning, even though the claim might
        be worded differently
14      -the context includes every semantic entity that is present in the claim
15      -the information in the context is unambiguous and its interpretation can only
        lead to the logical conclusions that are drawn in the claim
16   Search for the context, that fulfills all those criteria the best.
17
18   Input:
19   -text of a scientific paper
20   -claim generated by an LLM on the basis of this paper
21
22   Output:
23   -The extracted context
24
25   Answer with only the extracted sentences, exactly as they are consecutively! written
    in the document
26   You are not allowed to hallucinate or make any adjustments to the sentences. You are
    also now allowed to add additional comments or headlines.
27   Strictly reply with only those sentences.
28   Try to select as few sentences as possible. The response must include every
    information needed to fullfill the criteria, but should not contain any
    unnecessary information,
29   that is not needed to justify the claim.
30
31
32   Scientific Paper Text : "{documentText}"
33   Claim: "{answer}"

```

Listing A.9: LLM cited text span extraction prompt